A Discrete Choice Model with Segmentation and Embedding-Augmented Learning for Travel Behavior Analysis

Kwangho Baek[a], Alireza Khani[a,*]

[a] *University of Minnesota, Minneapolis, MN 55455, United States*

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ABSTRACT

Understanding travel behavior for establishing transportation policy has become increasingly complex as mobility options and data sources have diversified. This study introduces a Discrete Choice Model with Segmentation and Embedding-Augmented Learning (DCM-SEAL), bridging classical econometric models and modern data-driven machine learning. DCM-SEAL integrates three components within a single, end-to-end architecture: (1) a feed-forward deep neural network that identifies heterogeneous behavioral segments (latent classes) through nonlinear combinations of individual-specific variables; (2) class-specific embeddings that project high-cardinality categorical variables into a latent utility space to uncover behavioral relationships and within-class utility variations; and (3) a conventional linear-in-parameter utility formulation that preserves the direct interpretability grounded in random-utility theory. Synthetic data experiments confirmed that DCM-SEAL accurately recovers known behavioral parameters, demonstrating robustness and unbiased learning under controlled conditions. Complementary analyses using the SwissMetro benchmark data showed that the model achieved a test pseudo-R^2 of up to 0.482, exceeding several reported benchmarks under comparable SwissMetro settings while maintaining consistent value-of-time estimates and interpretable latent class structures. Beyond predictive performance, DCM-SEAL enhances the practicality of data-driven choice analysis. While individual machine learning techniques such as embedding and neural membership components alone improve either representation or segmentation, their joint integration within DCM-SEAL yields synergistic gains in deep behavioral understandings. Rather than presenting DCM-SEAL as a fully validated deployment tool, this paper positions it as a practically oriented modeling framework that unifies existing data-driven discrete choice model structures as special cases and provides a flexible basis for further applied transport policy.

1. Introduction

Travel mode and path choice analyses based on Discrete Choice Models (DCMs)—grounded in Random Utility Maximization (RUM) and typically adopting the Multinomial Logit (MNL) structure—have long been central to travel behavior research. Since the seminal work of McFadden (1974), the DCM framework has matured over five decades through ongoing efforts to overcome the statistical and behavioral limitations of early formulations. The MNL model remains widely used for its analytical tractability and ability to yield interpretable behavioral indicators such as the Value of Time (VoT) and elasticities. Yet, its reliance on independent and identically distributed (IID) error terms and linear-in-parameter utilities constrains flexibility, often reducing predictive accuracy or biasing estimates when the utility is not well specified.

Later developments have primarily aimed to relax these constraints by more effectively capturing taste heterogeneity, or variation in individual preferences. The Latent Class Choice Model or LCCM (Bhat, 1997; Kamakura and Russell, 1989) introduces a finite mixture structure that partitions travelers into behaviorally homogeneous latent segments, while the Mixed MNL or Random Parameter Model (McFadden and Train, 2000) assumes a continuous distribution of coefficients. Other advances incorporate latent psychological constructs through a structural-equation framework, forming Hybrid Choice Models (HCMs) such as the Integrated Choice and Latent Variable (ICLV) model (Ben-Akiva et al., 2002). These models link subjective factors, such as attitudes, perceptions, or beliefs, to observed choices via latent variables. In parallel, other extensions relax the IID assumption that conferred the name Generalized Extreme Value (GEV) model family, which allows correlation among similar alternatives, or Probit models, which adopt a fully flexible multivariate-normal error structure (Train, 2009). Some approaches even move beyond the RUM paradigm, drawing on alternative behavioral theories such as regret minimization (Chorus et al., 2008).

Despite their sophistication, even advanced theory-driven DCMs remain vulnerable to specification bias (van der Pol et al., 2014) and require analysts to manually define model structures like utility functions, variable selection, and interaction terms, based on prior behavioral assumptions (Rodrigues et al., 2022), which is often subjective and time-consuming. In contrast, data-driven DCMs have emerged as credible alternatives to mitigate these limitations (van Cranenburgh et al., 2022). Although early studies recognized the conceptual link between Artificial Neural Networks (ANNs) and DCMs (Bentz and Merunka, 2000; Hruschka et al., 2002), it is only recently—supported by advances in computing power and the rapid evolution of machine learning (ML) and artificial intelligence (AI)—that numerous ML-integrated DCMs have emerged, which is discussed in the next section. This shift has been further accelerated by the growing availability of large, complex, and passively collected transportation datasets, such as automated fare collection records (Wong and Farooq, 2021).

ML components in data-driven DCM offer capabilities beyond the reach of conventional econometric frameworks (Lee et al., 2018). *Deep neural networks* (DNNs), which are under the above ANN category where multiple hidden, intermediate layers of neurons are introduced, work as universal function approximators that can model highly non-linear relationships and implicit interactions directly from data without prespecified functional forms (Hornik et al., 1989). Moreover, ML introduces efficient data-representation techniques such as *embeddings* (Mikolov et al., 2013), which maps high-cardinality categorical variables into low-dimensional continuous vector spaces that capture latent structure and semantic similarity. By reducing dimensionality, embeddings help alleviate the curse of dimensionality (associated with the conventional dummy encoding) and mitigate the resulting overfitting by adopting parameter regularization, which are essential advantages for modeling with typical discrete choice datasets (Arkoudi et al., 2023; Biswal et al., 2025).

* Corresponding author
E-mail address: akhani@umn.edu (A. Khani).

Historically, government-funded large-scale travel surveys—such as household travel surveys and on-board transit surveys—are the cornerstone of regional and national transportation planning (Daly et al., 2007). Yet most data-driven DCMs to date have been evaluated primarily on synthetic datasets for proof-of-concept or on real-world choice datasets used for benchmarking predictive performance, such as the well-known SwissMetro dataset (Bierlaire et al., 2001), or being tested on both types of datasets (Han et al., 2022; Salvadé and Hillel, 2025; Sifringer et al., 2020; van Cranenburgh et al., 2022). Consequently, discussions of how these models can be deployed in real-world planning and transferability to another context remain limited (Ali et al., 2023).

This deployment gap is critical: models lacking flexibility and transferability often require significant structural changes when applied to contexts that differ substantially from their training data, such as cross-city policy transfers or long-horizon behavioral forecasts, thereby limiting their policy relevance (Feng et al., 2024). Practitioners, therefore, need a versatile data-driven DCM framework that produces diverse, nuanced, and behaviorally consistent insights with a structure that can encompass multiple modeling specifications, while remaining robust to variations in input data (Salvadé and Hillel, 2025). Such a framework would enhance the integration of ML-based behavioral modeling with traditional planning components beyond mode choice—extending to trip generation, distribution, and network assignment. For example, detailed behavioral segmentation derived from these models can directly support synthetic population generation within activity- or agent-based simulation frameworks (Kim and Mokhtarian, 2023), potentially aided by emerging deep generative models (Kim and Bansal, 2023).

Building on this motivation, we propose a modeling framework that is trainable on large-scale travel surveys and satisfies three key practical requirements. First, it should reduce the burden on practitioners by minimizing the need to manually specify model structures, such as utility function definitions or taste heterogeneity mechanisms, and by leveraging automatic learning capabilities of ML techniques. Second, it must preserve core econometric interpretability for key policy insights (e.g., VoT) while simultaneously providing rich, latent class or segment-specific, and context-dependent behavioral understandings for general applicability. Third, it should efficiently handle large numbers of features, particularly high-cardinality categorical variables prevalent in typical transportation planning source data, whose inclusion via traditional dummy encoding can lead to high dimensionality, overfitting, and multicollinearity in finite samples, hindering model optimization.

We term this framework a Discrete Choice Model with Segmentation and Embedding-Augmented Learning (DCM-SEAL), acting as an integrated platform that builds upon the traditional MNL model. While it incorporates the published Embedding-MNL (E-MNL) structure (Arkoudi et al., 2023), and goes beyond mere integration by formalizing the neural-membership LCCM—a concept previously confined to highly experimental or specific application contexts (Baek and Khani, 2025; Han, 2019)—into a synergistic, end-to-end structure. Here, conventional MNL and the two ML components (DNN, embedding) are unified within a single architecture in which input variables are assigned distinct roles based on either the analyst's informed choice or by a grid search. Each role follows a dedicated computational path that captures its specific contribution before converging into the conventional logit-based choice probability calculation stage. From an optimization perspective, these interconnected pathways are jointly updated via gradient backpropagation, enabling seamless integration into modern ML development frameworks.

Because all three components operate jointly in DCM-SEAL, multiple regularization methods are incorporated to ensure model stability, consistency, and practicability. Furthermore, since the assignment of variable roles can be subjective, DCM-SEAL offers two complementary strategies for practitioners: (1) a manual role configuration, grounded in the expected applications of model insights for transportation planning; and (2) a grid-search-based approach that exhaustively explores possible role combinations, each accompanied by automated hyperparameter tuning to identify optimal configurations.

From a planning perspective, the framework is intended for applications where analysts need both improved empirical fit and interpretable subgroup differences in sensitivity to time, cost, or other trip contexts. The contribution of this paper is therefore not direct policy evaluation itself, but a modeling structure that can support richer policy analysis while retaining planner-facing behavioral quantities.

In summary, the proposed DCM-SEAL framework seeks to bridge the methodological divide between theory-driven econometric models and data-driven ML approaches. While recent ML-integrated DCMs have achieved notable gains in predictive accuracy, they often do so at the expense of interpretability and practical applicability. By integrating neural segmentation and categorical embeddings within a logit-based utility structure, DCM-SEAL preserves the "core" behavioral transparency of classical models while extending their capacity to learn complex, non-linear relationships from large-scale survey data autonomously. Importantly, the framework serves as a generalized formulation that encompasses a spectrum of existing models—ranging from the conventional MNL or LCCM to the recent E-MNL and neural membership models introduced above—each of which represents a special case of DCM-SEAL. In doing so, this study contributes to the ongoing advancement of discrete choice modeling by proposing a framework that:

- Unifies interpretable core utility estimation, neural latent segmentation, and embedding-based representation learning within a single framework;
- Provides a modular variable-role architecture that nests MNL, E-MNL, and neural-membership LCCM as special cases; and
- Demonstrates, on synthetic and benchmark data, that the integrated framework can improve empirical fit while preserving interpretation of core behavioral parameters.

To encourage transparency and reproducibility in data-driven discrete choice modeling, the complete Python code, model architectures, and processing pipelines for the DCM-SEAL framework are made publicly available in our GitHub repository (https://github.com/UMN-Transit-Lab/BaekKhani_DCM-SEAL).

The remainder of this paper is organized as follows. **Section 2** reviews the relevant literature on integrating ML components with discrete choice models, and positions our proposed framework within recent methodological developments. **Section 3** details the architecture of DCM-SEAL, following the descriptions of its choice, segmentation, and embedding components. **Section 4** focuses on the synthetic dataset experiment, designed to test DCM-SEAL's ability to recover known behavioral parameters within a controlled setting. **Section 5** then applies DCM-SEAL to the public SwissMetro dataset for comparative result analyses to benchmark real-world performance against existing approaches. Finally, **Section 6** concludes with a summary of findings and directions for future research.

2. Literature Review

As noted by van Cranenburgh et al. (2022), research integrating ML techniques with traditional DCMs primarily seeks to overcome the limitations of the theory-driven modeling paradigm. These efforts mainly target two challenges: (1) mitigating specification bias by allowing models to autonomously learn complex utility structures (including non-linearities and high-order interactions) to reduce endogeneity, and (2) enhancing the ability to capture both systematic and random taste heterogeneity. Within this growing literature, we classified data-driven DCM studies into three methodological streams that align with the proposed DCM-SEAL framework: (1) enhancing the flexibility of utility functions while preserving interpretability; (2) employing neural networks for segmentation and modeling taste heterogeneity; and (3) implementing embeddings for feature representation learning.

Regarding the first stream, traditional DCMs require analysts to manually define complex model structures, often relying on *a priori* behavioral assumptions (Rodrigues et al., 2022; Salvadé and Hillel, 2025). When the utility specification is incorrect, the model may produce biased estimates, misleading behavioral interpretations, and poor predictive performance (Torres et al., 2011). The conceptual connection between DNNs and DCMs to resolve the issues was first recognized in the seminal work of Bentz and Merunka (2000), who showed that a simple perceptron (a shallow neural network with log-linear or *SoftMax* outputs) is mathematically equivalent to the MNL model. In this sense, an MNL formulation can be viewed as a special case of a DNN, corresponding to a one-layer network with SoftMax activation and no regularization. Furthermore, the model fitting optimization objectives for both models, maximum likelihood estimation in MNL and cross-entropy minimization in DNN, are mathematically equivalent (Sifringer et al., 2020) under basic training conditions where no regularization techniques are enforced.

Motivated by these findings and supported by advances in computational power, researchers have developed numerous data-driven DCMs for transportation-related choice problems. Instead of replacing the entire MNL structure with a fully connected DNN, as earlier studies attempted, Wang et al. (2020) proposed the *DNN with Alternative-Specific Utility (ASU-DNN)* model, which separates input features for different alternatives so that each alternative is processed through its own network. Individual-specific features are subsequently combined to compute the final utility, representing a strategic use of transportation domain knowledge to constrain the architecture and enhance interpretability. Wong and Farooq (2021) introduced *ResLogit*, which integrates the *ResNet* architecture (He et al., 2016) into the logit utility function. By incorporating skipped connections and residual layers, this residual function captures non-linear cross-effects and unobserved choice heterogeneity without requiring explicit prior specification, thereby substantially reducing specification bias. In parallel, Wang et al. (2021) was also motivated by *ResNet* and proposed a mix of data- and theory-driven DCMs, named *TB-ResNet*, via a weighted sum of outputs from both components (i.e., a model ensemble). It should also be noted that DNNs are not the only tool for flexible utility specification. Rodrigues et al. (2022) proposed *MNL-ARD (Automatic Relevance Determination)* using a Bayesian framework to automatically determine relevant variables, while Salvadé and Hillel (2025) developed RUMBoost, which replaces every parameter in the RUM model with an ensemble of gradient-boosted decision trees.

However, reinforcing the MNL structure with more flexible or lower-bias components like the above studies, although often improving predictive accuracy, comes at the cost of reduced interpretability, reflecting the well-known bias-variance trade-off (Hastie et al., 2009). This interpretability loss can result in so-called “black box” models, which are particularly problematic in transportation research where credible policy implications depend on economically meaningful indicators, such as negative choice elasticities with respect to cost or travel time. In addition, solely relying on data-driven components can exacerbate an inequality in the data structure, raising ethical challenges (Zheng et al., 2021).

To balance flexibility and interpretability, several studies have adopted hybrid modeling strategies that combine theory-driven and data-driven components within a single framework. The *Learning Multinomial Logit* (*L-MNL*) proposed by Sifringer et al. (2020) divides the systematic utility into two parts: a knowledge-driven, interpretable component structured as a conventional linear-in-parameters model, such as MNL, and an additive, data-driven representation term modeled by a DNN. To preserve interpretability, the two parts use disjoint sets of input variables, ensuring that each variable contributes either to the interpretable or to the learned component, but not both. A related approach is the *TasteNet-MNL* model by Han et al. (2022), which integrates a neural network (*TasteNet*) to represent systematic taste heterogeneity. Unlike L-MNL, TasteNet does not introduce a single additive DNN-derived term; instead, their neural network outputs a set of coefficients, or *taste parameters*, as flexible functions of individual characteristics. These parameters are then multiplied by specific alternative attributes, such as travel time or cost, to form the utility. Both models require analysts to manually specify which variables remain directly interpretable, which are treated as flexible (i.e., processed by the DNN), and, for TasteNet, additionally require a specification of which ones interact with these flexible tastes.

Another challenge in adopting advanced hybridization techniques in DCMs is ensuring that estimated parameters retain behavioral validity, particularly regarding the sign and monotonicity of utility coefficients. To address this issue, several studies have introduced explicit domain-knowledge constraints within data-driven DCMs. For instance, similar to how theory-driven models specify parameter distributions to impose behavioral consistency (Fosgerau, 2006), the aforementioned TasteNet-MNL model applies sign constraints to DNN outputs to preserve monotonic relationships in individual taste heterogeneity. Other approaches include DCM-LN by Kim and Bansal (2024), which employs lattice networks (LN) with monotonicity constraints to ensure predictable attribute effects, and the gradient regularization technique by Feng et al. (2024), which constrains the direction of gradients during training to maintain demand monotonicity. Collectively, these methods function as domain-informed regularization mechanisms, preventing overfitting, avoiding implausible attribute dynamics that affect the resulting choice probability, and enhancing interpretability.

The second group of data-driven DCMs we analyzed employs DNN to enhance segmentation within the LCCM framework. The conceptual compatibility between DNN classifiers and LCCMs arises from the fact that the LCCM’s membership function is mathematically equivalent to a feedforward neural network without a hidden layer (Vermunt and Magidson, 2003). Despite this connection, neural-membership LCCMs are not yet well-established in the transportation literature and remain largely experimental. Among the few examples, Han (2019) provided an initial proof-of-concept exploring technical possibilities of replacing the traditional shallow logit membership function with a DNN, while Baek and Khani (2025) focused this neural extension on a highly specific planning application. Our proposed framework unifies these isolated, experimental endeavours to yield synergistic improvements in predictive accuracy, interpretability, and policy relevance. In another related effort, Lahoz et al. (2023) proposed a DNN-integrated hybrid choice model in which DNNs process attitudinal survey data to create latent variables, assuming that different latent classes respond differently to these variables. However, the DNN itself was not used to identify class membership. Notably, several leading studies focusing on utility flexibility or feature representation have explicitly recognized the integration of LCCM structures as a promising direction for future research (Arkoudi et al., 2023; Rodrigues et al., 2022; Sifringer et al., 2020).

Returning to higher-level modeling choices, LCCMs and other finite mixture models—alongside their continuous counterparts, such as random parameter MNLs—are widely recognized for their ability to capture taste heterogeneity, wherein individuals exhibit diverse travel behaviors even under similar conditions. Numerous transportation studies have explored this behavioral variability, including heterogeneous perceptions of transit transfers (Navarrete and Ortúzar, 2013) and platform waiting time (Fan et al., 2016), as well as heterogeneity in technology adoption behaviors, such as the use of real-time information (Watkins et al., 2011). In addition to the fact that the LCCM membership function can be seamlessly substituted by and enhanced through DNNs, Kim and Mokhtarian (2023) highlights several conceptual reasons for preferring LCCMs over random parameter models when modeling taste heterogeneity:

- It is cognitively more manageable to interpret parameters of a small number of discrete groups rather than continuous distributions.
- Unlike random parameter models, where the correspondence between individuals and parameter values is uncertain, LCCMs assign each individual a fixed membership probability, with members of the same class sharing fixed parameter values.
- Between-class interpretation is possible in LCCMs; variables contributing to utility in one class may be insignificant in another, offering richer behavioral insights
- Class-specific profiling is feasible, enabling researchers to compute and compare behavioral summaries for each latent group.

Another important modeling consideration in adopting the LCCM framework is determining the number of latent classes. As noted by Wagenmakers et al. (2012), class numbers may be either predefined or estimated heuristically. The former corresponds to a confirmatory approach (Laudy et al., 2005), whereas the latter represents an exploratory approach. Early mixture models primarily relied on exploratory estimation (Hoijtink, 2001) due to its practical convenience and the perception that confirmatory modeling requires extensive domain knowledge (Hess and Daly, 2014). Nevertheless, confirmatory designs can be valuable for testing specific hypotheses by deliberately partitioning samples and applying different model specifications (i.e., distinct utility functions) across groups (Anowar et al., 2019). In practice, this often involves first estimating exploratory LCCMs separately for each subgroup—ideally with the same number of classes—and then comparing class-specific parameter estimates to evaluate how behavior differs across groups (Olaru et al., 2011). This two-step approach effectively combines domain knowledge from confirmatory modeling with the unsupervised classification capacity of exploratory models.

On the other hand, several statistical and practical factors could also come into play when selecting the optimal number of classes. Nylund et al. (2007), using Monte Carlo simulation, found that models with the smallest Bayesian Information Criterion (BIC) values generally perform best. However, Grimm et al. (2017) emphasized that there is no universally accepted criterion for class-number selection. In practice, analysts often prefer interpretable models with fewer latent classes, even if models with more classes achieve slightly higher accuracy (Mouter et al., 2017). Moreover, classical asymptotic criteria such as BIC are not directly applicable to models incorporating ML components, since their effective degrees of freedom are unknown (van Cranenburgh et al., 2022).

The third stream of data-driven DCMs focuses on feature representation learning through embeddings. Within the transportation domain, only a few studies have employed embeddings in discrete choice modeling while explicitly emphasizing interpretability. Arkoudi et al. (2023) proposed the *Embedding-MNL* (*E-MNL*) model, which embeds all unique levels of categorical variables for each output term to represent the variable's specific contribution to each choice alternative. This approach substantially reduces the dimensionality of the utility formulation while preserving interpretability, as the embedding weight matrix itself can be analyzed to identify more influential category levels using cosine similarities. The authors argue that this structure not only improves parsimony but also enhances behavioral insight. Beyond transportation, Wang et al. (2023) incorporated embeddings into a Transformer-DCM framework for multi-choice prediction in consumer behavior studies. Similarly, although not within a discrete choice context, Biswal et al. (2025) applied embeddings to handle high-cardinality input data, resulting in higher model accuracy, also motivated by a similar need to reduce dimensionality as in Arkoudi et al. (2023). However, unlike E-MNL, their embedding dimensions were flexible and not tied to each alternative's utility space, as the method was designed for a regression task. They further found that passing embedding outputs through an additional activation layer introduced a non-linear transformation that significantly improved predictive accuracy, though such a configuration is less applicable to DCMs, where an alternative specific behavioral interpretation is essential.

The studies reviewed above show that recent ML-integrated DCMs have advanced along several related but distinct directions. Some studies use neural networks to relax or diagnose utility specifications; others preserve interpretable core parameters while adding data-driven representation terms; still others focus on taste heterogeneity, domain-informed DNN architectures, automatic utility specification, or embedding-based categorical representation. The purpose of this review is not to claim that neural networks, embeddings, or latent-class structures are individually novel, but to identify which modeling needs have been jointly or separately addressed in the existing literature.

The literature review indicates that the depth and maturity of research across the three methodological streams relevant to DCM-SEAL vary considerably. Numerous studies have explored replacing, augmenting, or constraining the MNL structure with neural networks or other ML architectures, whereas fewer have examined embedding-based representation within discrete choice models. Likewise, DNN-based membership modeling within LCCMs remains relatively underdeveloped despite the rich literature on latent class models and their relevance for interpretable behavioral segmentation (Kim and Mokhtarian, 2023). These gaps are important because practical travel behavior applications often require more than predictive improvement alone: they require interpretable core behavioral parameters, meaningful segmentation, and a tractable way to incorporate categorical variables without excessive manual specification.

Among these issues, determining the number of latent classes remains a central modeling decision for any LCCM-based framework, including DCM-SEAL. While this choice has long involved trade-offs between model fit and interpretability, its complexity is magnified when LCCMs are integrated with ML components. The literature suggests that some degree of exploration is still required in practice, but the number of classes is often bounded by practical interpretability and the analyst's purpose. In other words, selecting the optimal number of latent classes remains partly heuristic. This consideration motivates the design of DCM-SEAL as a flexible framework in which class membership, class-specific utility parameters, and categorical embedding terms can be explored jointly while preserving direct behavioral interpretation for core variables.

3. Theory, The Model, and Training

This section outlines the theoretical foundation and architecture of the proposed DCM-SEAL model. The first three subsections describe its three core components—DCM with a latent-class structure, DNN for latent class membership, and embedding integration for categorical variable representation. **Section 3.4** explains how these components are unified within a single end-to-end structure, discusses DCM-SEAL properties, and then describes the hyperparameter training strategy for model deployment in the last subsection.

3.1. Discrete Choice Model and Latent Class Structure

Since the incorporation of RUM theory into econometric modeling, the DCM framework has become indispensable for analyzing travel behavior and informing transportation planning (McFadden, 2001). Over time, researchers have expanded DCMs to incorporate diverse behavioral assumptions that extend beyond strict utility maximization, as summarized by Ramos et al. (2014). Despite these conceptual advances, the functional form of most DCMs has remained closely related to the closed-form logit formulation, also known as SoftMax in machine learning literature, due to its analytical simplicity and broad applicability. Consequently, model variants are often referred to as specific forms of "logit" models, like the *Nested Logit* (Ben-Akiva and Lerman, 1985). The general MNL formulation is expressed as:

$$P_{ni} = \frac{\exp(V_{ni})}{\sum_{j\in \boldsymbol{J}} \exp(V_{nj})} \tag{1}$$

Equation (1) represents the choice probability P_{ni} of individual n selecting alternative i from their choice set $\boldsymbol{J}$. The systematic utility for each alternative i is typically defined as the linear-in-parameter form $V_{ni} = \boldsymbol{\beta}^T \mathbf{x}_n$, where $\boldsymbol{\beta}$ is the vector of estimated parameters associated with independent variables $\mathbf{x}_n$, such as travel time, cost, or categorical demographic indicators. The true utility U_{ni} is defined as the sum of the systematic component V_{ni} and a random error term ε_{ni}, which is assumed to follow an IID Gumbel distribution, leading to the choice probability form in **Equation (1)**. However, these estimates generally fail to capture taste heterogeneity among individuals, as they represent only point estimates for key travel-related variables such as time and cost.

Alternatively, categorical individual characteristics (i.e., unique levels) can be assigned their own parameter via one-hot (dummy) encoding, but this only isolates the marginal contribution of a binary subgroup (e.g., male versus non-male) when incorporated into the utility formula. Moreover, it restricts behavioral variability to fixed coefficients and overlooks richer sources of heterogeneity. Finally, including more individual-specific categorical variables or expanding the set of available alternatives often undermines model parsimony, leading to high dimensionality and potential overfitting.

To address these limitations, studies have extended the DCM framework by incorporating mixture modeling, as outlined by Train (2009). Mixture modeling represents individual heterogeneity as either a finite or an infinite (continuous) mixture of utility functions, depending on whether the distribution of tastes is assumed to be discrete or continuous. In the finite mixture or LCCM case, the choice probability P_{ni} can be expressed in a generalized form as:

$$P_{ni}: f(i|\mathbf{x}_n, \mathbf{s}_n) = \sum_{c=1}^{C} \Pr(c|\mathbf{s}_n) f_c(i|c, \mathbf{x}_n) \tag{2}$$

In **Equation (2)**, $f(i)$ denotes the probability mass function of outcome i. The vector $\mathbf{x}_n$ represents the covariates or a set of input variables for individual n, and the other vector $\mathbf{s}_n$ denotes another input set, segmentation bases—the individual-specific variables that determine latent class membership $c \in \{1,2, \dots C\}$. Accordingly, $\Pr(c|\mathbf{s}_n)$ defines the *class membership model*, and $f_c(i|c, \mathbf{x}_n)$ describes the *class-specific choice probability* or decision rule, mostly following the MNL form like **Equation (1)** marginally. In essence, finite mixture models or LCCMs partition the population into subgroups that are homogeneous within classes but heterogeneous across classes in their behavioral patterns (Bhat, 1997).

3.2. Deep Neural Network for Latent Class Membership Model

Classification problems are where machine learning techniques are popularly employed to predict a categorical label for a given input. Formally, classification is optimizing a function or classifier $g: \mathbb{R}^S \mapsto \mathcal{C}_c \in \mathbb{C}$, where S represents the one-hot encoded input feature dimension of the segmentation bases $\mathbf{s}_n$, and $\mathbb{C} = \{\mathcal{C}_1, \dots, \mathcal{C}_C\}$ denotes the set of possible classes with C distinct outcomes. It is assumed that there exists a fixed, unknown distribution over $\mathbb{R}^S \times \mathbb{C}$ where dataset $\mathbb{D} = \{(\mathbf{s}_1, c_1), \dots, (\mathbf{s}_N, c_N)\}$ of N samples—with the n-th sample belonging to the known class c_n—is drawn from. The goal is to find a classifier g that minimizes the loss $L(g) = P(g(\mathbf{x}) \neq c); c \in \mathbb{C}$ over all samples.

ANNs represent a class of machine learning classifiers inspired by the functioning of the human brain (Bishop, 2006). These networks consist of layers of interconnected neurons that process inputs and iteratively adjust their internal weights to make accurate predictions. ANNs excel at learning complex feature representations from raw data by automatically identifying relevant patterns and hierarchical relationships among features. By increasing the number of hidden layers, the model becomes a deep ANN (DNN), which can effectively capture non-linear decision boundaries and adapt to diverse data distributions without relying on predefined functional assumptions.

A feed-forward, fully connected DNN denoted $\Lambda(\cdot)$, and consisting of $A > 1$ hidden layers, is defined as shown in **Equation (3)**. Here, $\phi(\cdot)$ denotes the vector-valued activation function, typically set to the Rectified Linear Unit (ReLU). $\mathbf{W}_a$ denotes the weight matrix linking $(a-1)$-th and a-th layers, and $\mathbf{b}_a$ the bias vector for the a-th hidden layer. The weights and biases are optimized according to **Equation (4)**, where $\Omega(\mathbf{W})$ denotes the regularization term. In this formulation, $\hat{\pi}_n^c$ denotes the estimated probability of sample n belonging to class c, $\mathcal{L}'(\cdot)$ is the cross-entropy loss $-\mathbb{E}_n[\boldsymbol{\pi}_n \cdot \log\hat{\boldsymbol{\pi}}_n]$, and $\|\cdot\|_2$ is the L2-norm to regularize weights $\mathbf{W}$ with hyperparameter λ. Finally, $\mathbf{\Pi}$ is a binary indicator matrix representing all samples' class memberships derived from the observed data or ground truth, serving as the reference for loss computation.

$$\Lambda(\mathbf{s}_n; \mathbf{W}, \mathbf{b}) = \hat{\boldsymbol{\pi}}_n = [\hat{\pi}_n^1, \hat{\pi}_n^2, \hat{\pi}_n^3, \dots, \hat{\pi}_n^C] = \mathrm{SoftMax}(\mathbf{W}_{A+1}^T \phi(\mathbf{W}_A^T(\phi \dots \phi(\mathbf{W}_1^T \mathbf{s}_n + \mathbf{b}_1) \dots) + \mathbf{b}_A) + \mathbf{b}_{A+1}) \tag{3}$$

$$Min\ L: \mathcal{L}'(\hat{\mathbf{\Pi}}, \mathbf{\Pi}) + \Omega(\mathbf{W}) = -\frac{1}{N}\sum_{n=1}^{N}\sum_{c=1}^{C} \pi_{nc} \log(\hat{\pi}_{nc}) + \lambda \sum_{a=1}^{A+1} \|\mathbf{W}_a\|_2^2 \tag{4}$$

In modern practice, stochastic gradient-based optimization algorithms such as the *Adam* optimizer (Kingma and Ba, 2015) are widely employed. Adam inherently includes L2 regularization as a built-in feature called *weight decay*, which adapts to each parameter based on its gradient and the moving averages of past gradients. Consequently, it is often more appropriate to denote this regularization compactly as $\Omega(\mathbf{W})$ or $\Omega(\mathbf{W}; \lambda)$ when using such optimizers, rather than writing the regularization explicitly as the additive term, as in the last term of **Equation (4)**. Furthermore, a decoupled variant of Adam, known as *AdamW* (Loshchilov and Hutter, 2019), was later introduced to address inconsistencies caused by Adam's entanglement of weight decay with gradient updates. AdamW separates these two processes by directly applying weight decay to the parameters, leading to more consistent regularization and improved generalization performance.

In addition to weight decay, current practice for DNN regularization is employing dropout and mini-batching. Dropout randomly deactivates a subset of neurons during training, preventing co-adaptation among units and promoting robustness in the learned representations. This stochastic omission effectively averages over multiple network configurations, thereby reducing overfitting. Meanwhile, mini-batching introduces implicit regularization by estimating gradients from small, randomly sampled subsets of the data rather than the full dataset. The resulting gradient noise discourages convergence to sharp minima and improves the model's generalization capability. Together, these three mechanisms—AdamW's decoupled weight decay, dropout, and mini-batch training—jointly constrain overfitting while maintaining efficient and stable learning dynamics.

This structure generates the following seven hyperparameters, which must be tuned during model development. Hyperparameter tuning refers to optimizing model-specific settings that are not directly learned during training (discussed in detail in **Section 3.5**):

- A: the number of fully connected hidden layers in the network.
- N_a: the number of artificial neurons in the a-th hidden layer.
- γ: the learning rate, or step size, for each parameter update or epoch.
- N_E: the number of training iterations, or epochs, in the optimization of weights and biases.
- N_B: the number of mini-batches in each epoch used for incremental gradient updates.
- p_d: the dropout probability, representing the fraction of neurons randomly deactivated ("zeroed") at each training step.
- λ_s: the strength of weight decay regularization, where 0 indicates no decay and values approaching 1 push weights toward zero.

In summary, the DNN component of the proposed DCM-SEAL, denoted as $\Lambda(\mathbf{S}; \mathbf{W}, \mathbf{b})$, receives the all-sample segmentation bases $\mathbf{S} \in \mathbb{R}^{N \times S}$ as input and outputs class membership probabilities $\hat{\mathbf{\Pi}}$. Its structure is determined by the number of hidden layers A and the number of neurons N_a in each layer $a \in \{1,2,3, \dots, A\}$. Regularizations $\Omega(\mathbf{W}; p_d, \lambda_s)$ are applied throughout N_E epochs, governed by the learning rate γ and mini-batch size N_B. The resulting "trained" network serves as the broadcasted latent class membership function $P(\mathbf{c}|\mathbf{S})$ from **Equation (2)**. Details on how Λ is incorporated into the complete DCM-SEAL framework, and the procedure for hyperparameter tuning is discussed in **Section 3.5**.

3.3. Embedding Integration in DCM

The concept of embedding was initially developed to learn a compact, continuous representation of discrete tokens, such as words, by mapping a vocabulary of millions of items into a tractable vector space, typically with 50 to 300 dimensions (Mikolov et al., 2013). This approach assumes an input represented as a one-hot (or dummy-encoded) categorical vector $\mathbf{e}$, where a variable with Z possible levels is encoded into a Z-dimensional binary vector containing a single 1 at the index corresponding to the observed category and 0 elsewhere. An embedding layer replaces this sparse, high-dimensional representation with a dense vector by training a learnable weight matrix $\mathbf{M} \in \mathbb{R}^{Z\times J}$. The embedded output vector $\mathbf{v} \in \mathbb{R}^{J}$ is then obtained by the matrix multiplication $\mathbf{v} = \mathbf{e}^T\mathbf{M}$, where J denotes the embedding (output) dimension. Although J is arbitrary and typically much smaller than Z, sufficient training data and an appropriate higher-level objective (e.g., in a transformer-based large language model trained on texts from the entire Internet) allow each dimension or combinations of dimensions to capture meaningful latent structures.

For instance, in natural language processing, word embeddings can demonstrate and enable a "mathematical operation" of semantic meanings via vector arithmetic, like $\mathbf{v}_{\text{King}} + (\mathbf{v}_{\text{Woman}} - \mathbf{v}_{\text{Man}}) \approx \mathbf{v}_{\text{Queen}}$, where the difference between the embedded vectors of "woman" and "man" represents a learned gender direction in the latent space. In this sense, an arbitrary direction like $\mathbf{v}_{\text{Woman}} - \mathbf{v}_{\text{Man}}$ in the embedding space can signify interpretable relations (i.e., "feminine" direction in this example) that are not apparent in the original, high-dimensional one-hot categorical space. On the other hand, modelers can also train individual embedding dimensions themselves to align with meaningful latent concepts.

Focusing on this latter approach, Arkoudi et al. (2023)'s E-MNL successfully mapped the unique levels of the surveyed categorical variables into an embedded space where each axis corresponds to the utility contribution of a choice alternative in the DCM. They formulated the systematic utility of alternative i for individual n as the sum of linear-in-parameter terms for both alternative-specific variables x_{nk}^i ($k \in \{1,2,3,\ldots,K\}$) and embedded categorical characteristics x_{ne}^i ($e \in \{1,2,3,\ldots,E\}$), as shown in **Equations (5)** and **(6)**. The sets of alternative-specific and categorical variables are kept disjoint to preserve the direct interpretability of the marginal rate of substitution derivable between a β_k's pair.

$$V_{ni} = \sum_{k=1}^{K} \beta_k x_{nk}^i + \sum_{e=1}^{E} \beta_e^i x_{ne}^i \tag{5}$$

$$x_{ne}^i = (\mathbf{e}_{ne}^T \mathbf{M}_e)[i] \tag{6}$$

Here, $\mathbf{e}_{ne}$ denotes the one-hot encoded vector representation of the e-th categorical variable for individual n (length Z_e), and $\mathbf{M}_e$ is the trained embedding weight matrix mapping a vector of length Z_e to a vector of length J (the number of alternatives). The operation $[i]$ returns the i-th element of the preceding row vector. For compact representation, all categorical variables $\mathbf{e}_{ne}$ of varying e can be concatenated to form a longer vector $\mathbf{e}_n$, that we named *embedding bases*, with length $Z = \sum_{e=1}^{E} Z_e$. Using these embedding bases, an embedding input matrix storing all categorical levels for each alternative can be constructed through an operation we named *selective replication*, denoted $R(\mathbf{e}_n)$. This operation duplicates $\mathbf{e}_n$ by E times to create a matrix of size $E \times Z$, retaining only the column entries corresponding to the active category (otherwise all zeroed out) in each row representing e-th categorical variable.

Figure 1 illustrates this structure and process using an example with two categorical variables ($E = 2$): *Gender* ($e = 1$) and *Income* ($e = 2$). Gender has three levels ({M: Male, F: Female, O: Others}; $Z_1 = 3$) and Income has 2 levels ({H: High, L: Low}; $Z_2 = 2$), and the choice set includes three alternatives—Car, Bus, and Metro ($J = 3$). Under this configuration, a single embedding weight matrix $\mathbf{M}$ of size $Z \times J$ (where $Z = Z_1 + Z_2$) simultaneously embeds all categorical variables and outputs a matrix of x_{ne}^i values with dimensions $E \times J$. These embedded values are then used to compute the second term in **Equation (5)**, which captures the contribution of categorical variables to the systematic utility V_{ni}.

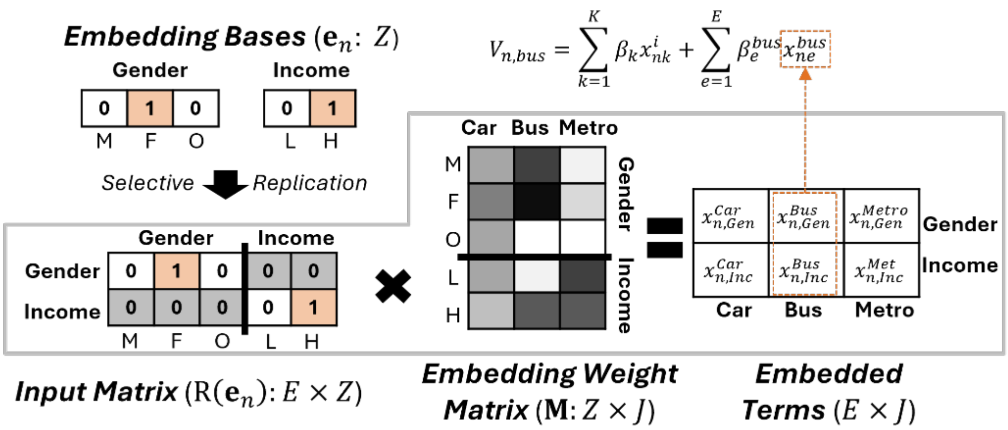

Figure 1 Example of embedding generation for categorical variables

To ensure interpretability, E-MNL introduced two key constraints: (1) each row of embedding weight matrix $\mathbf{M}$ is L2-normalized so that its norm equals 1, and (2) the signs of the estimated parameters β_e^i are constrained to be non-negative. Together, these constraints preserve behavioral consistency and interpretability. The first guarantees that each categorical level contributes proportionally to generating embedded terms within a bounded range (i.e., the remaining contribution is addressed by β_e^i), while the second ensures that the effects of categorical embedding remain directionally consistent for interpretation. As a result, the cosine similarity between a pair of rows of the embedding matrix can be used to assess correlations among categorical levels in terms of their utility contributions (see **Section 5.2.2** for more details).

DCM-SEAL adopts the embedding component in a manner like that of E-MNL, with an extension to regulate how embeddings are treated across latent classes, since the original E-MNL does not include a class structure. Additional regularization may also be applied to the embedding matrix to mitigate overfitting, but no new hyperparameter types are introduced beyond those already defined in **Section 3.2**. This is because the embedding component is seamlessly optimized within DNN's same iterative training process using AdamW, with the inclusion of a single additional weight decay hyperparameter λ_e applied to the embedding weight matrix (i.e., different from λ_s for the DNN component).

3.4. DCM-SEAL Structure: Putting the Three Components Altogether

The proposed DCM-SEAL unifies the three core components described in **Sections 3.1** through **3.3**—the latent class structure, DNN membership function, and the embedding-based representation—into a single, end-to-end differentiable architecture. This unified framework enables all components to be jointly optimized via gradient backpropagation while maintaining clear econometric interpretability for key behavioral parameters. The integration is achieved by assigning each explanatory variable in the dataset a distinct, non-overlapping role that determines its computational pathway within the model.

3.4.1. The Trinity of Variable Roles

Given a choice dataset $\mathbb{D} \in \mathbb{R}^{N\times(D+1)}$, where the last column (+1) stores the observed choice indices $\mathbf{y}_0$, the modeling process begins by classifying the D explanatory variables into *Core* and *Non-Core* groups. Core variables, such as in-vehicle time (IVT), out-of-vehicle time (OVT), and fare, represent quantities whose coefficients have direct behavioral meaning, such as marginal rates of substitution (MRS), willingness-to-pay (WTP), or VoT. From a technical perspective, explanatory variables are further distinguished as categorical or numerical. Each variable is then assigned to one of three mutually exclusive roles that define how it interacts with the model's computational layers:

- **Linear-in-Parameter Variables (LIP: $\mathbf{X}_n$'s)**: A total of K_0 raw variables are directly and linearly incorporated into the systematic utility of each class. Their class-specific coefficients β_k^c are estimated separately by class yet retain the same behavioral interpretation as in a conventional MNL model. Core variables are always included in this role, optionally supplemented by individual-specific characteristics needed for MRS calculations. Both numerical and categorical variables can serve as LIP variables, but categorical ones must first be encoded[1] before obtaining identifiable coefficients β. The final post-encoding form of the LIP variables, $\mathbf{X}_n$, takes the shape of a matrix in which each row corresponds to an alternative. The number of resulting columns denoted K, represents the total count of variables after encoding, including those for alternative-specific constants (ASCs).
- **Segmentation Bases (SB: $\mathbf{s}_n$'s)**: A total of S_0 raw variables are used as segmentation bases and processed by the DNN component (Λ) to produce class-membership probabilities $\hat{\pi}_n^c$'s. This module captures non-linear and interaction effects—particularly among demographic and contextual variables—that influence the likelihood of belonging to each latent class. Both numerical and categorical variables are acceptable: the former are standardized (zero mean, unit variance), while the latter undergo one-hot encoding to fit the DNN architecture. The post-encoding dimension of the segmentation input $\mathbf{s}_n$ to DCM-SEAL is denoted S.
- **Embedding Bases (EB: $\mathbf{e}_n$'s)**: A total of E remaining raw non-core categorical variables not assigned to LIP or SB roles are automatically classified as embedding bases. Each is represented as a compact, continuous vector in a (class-specific) embedding space. This mechanism replaces traditional one-hot encoding and enables alternative-specific and optionally class-specific behavioral interpretation while alleviating dimensionality issues. As discussed in **Section 3.3**, the selected variables are first encoded to produce length-Z vectors $\mathbf{e}_n$, which serve as each individual's embedding input for DCM-SEAL.

As illustrated in **Figure 2**, the selection of variable roles is influenced by both modeling objectives and the technical nature of the data. The complete set of variables must satisfy exhaustiveness[2] and mutual exclusiveness, expressed as below **Equation (7)**. This condition ensures that each variable contributes through a single, well-defined behavioral pathway, thereby avoiding confounding across roles.

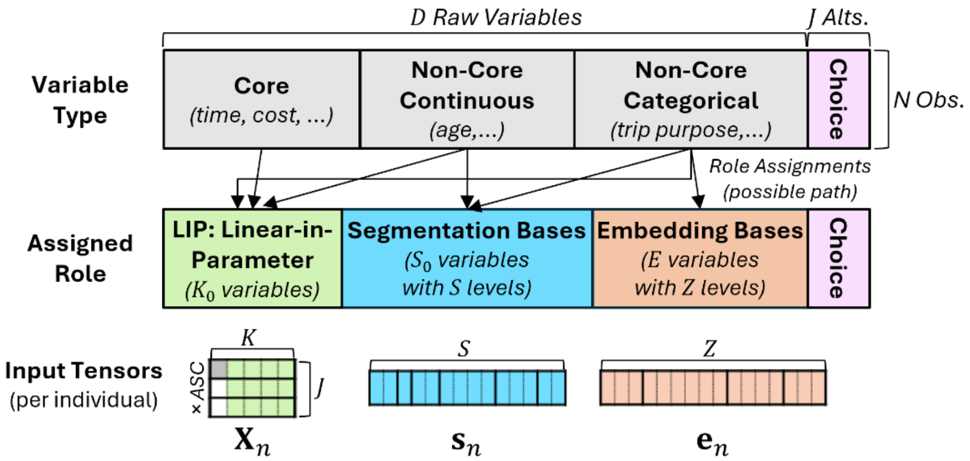

Figure 2 Possible variable role assignments by variable types and corresponding DCM-SEAL inputs

$$\mathbb{D}\backslash\mathbf{y}_0 = \mathbf{LIP} \cup \mathbf{SB} \cup \mathbf{EB}, \textit{where } \mathbf{LIP} \cap \mathbf{SB} = \mathbf{LIP} \cap \mathbf{EB} = \mathbf{SB} \cap \mathbf{EB} = \emptyset \text{ and } D = K_0 + S_0 + E \tag{7}$$

In practice, DCM-SEAL supports two strategies for determining variable roles:

- Manual configuration, grounded in behavioral reasoning for transportation planning and downstream applications.
- Grid-search exploration, which systematically tests alternative role combinations.

[1] This encoding follows the conventional treatment of alternative-specific constants (ASCs) and categorical variables in the MNL utility specification. For a dataset with a maximum of J alternatives, the input vector $\mathbf{x}_n$ is first replicated J times row-wise, after which a column of dummy indicators (a vector of ones) is concatenated to represent the $J-1$ ASCs, with one alternative set as the reference.

[2] In conventional theory-driven DCMs and many parametric econometric models, exhaustive inclusion of all available features is generally discouraged for reasons of model parsimony and multicollinearity. In contrast, machine learning models often benefit from including all candidate inputs, even when they are correlated or partially redundant. This robustness stems from the intrinsic capacity of modern architectures (e.g., ReLU-based DNNs) and regularization techniques (such as weight decay or dropout) to automatically attenuate redundant features during training.

Both approaches begin by fixing the LIP variables, primarily drawn from the core variables that form the behavioral focus of the modeling task. Among the remaining non-core (peripheral) variables, continuous variables are first assigned to SB, while the remaining categorical variables are candidates for either SB or EB.

Under the manual configuration approach, socio-demographic or individual attributes (e.g., gender, age, income, household type) can be designated as segmentation bases, and trip-contextual variables (e.g., time of day, day of week) can be assigned as embedding bases. This division aligns with the conventional segmentation practice, where Kim and Mokhtarian (2023) found in their review that 72% of LCCM studies employed demographic traits as segmentation variables, and it also enhances compatibility with activity-based or agent-based modeling (ABM) frameworks (Wafa et al., 2015), which has now become a cornerstone of general transportation planning. Specifically, in recent simulation-based ABMs, the initial stage typically involves synthesizing a diverse population across demographic and geographic segments using available datasets, followed by resampling procedures to reflect the target simulation context (Borysov et al., 2019; Kim and Bansal, 2023). Accordingly, this type of manual configuration strategically assigns variables helpful in defining and applying population synthesis—namely, demographic factors or features tied to individuals—to SB, while variables that represent contextual or scenario-specific variation are allocated to the EB.

Conversely, the grid-search approach imposes no *a priori* assumptions about the appropriate roles of variables. The search systematically explores all possible partitions, evaluating each configuration based on quantitative model-fit statistics and, if applicable, qualitative assessments of how effectively taste heterogeneity is captured. The final configuration is selected to achieve an optimal balance between behavioral interpretability and empirical performance.

3.4.2. A Unified Computational Flow for Utility Formulation

DCM-SEAL architecture computes choice probabilities through a structured sequence of operations that jointly incorporate segmentation, embedding, and class-specific linear utilities. First, the class-agnostic representative utility in **Equation (5)**, V_{ni}, is extended into class-specific utility V_{ni}^c, as shown in **Equation (8)**. This introduces class-dependent parameters for both LIP variables (β_k^c) and EB terms (β_e^{ci}), together forming the class-specific learnable parameter vector $\boldsymbol{\beta}^c = \{\beta_k^c|\forall k\} \cup \{\beta_e^{ci}|\forall e\}$:

$$V_{ni}^c = \sum_{k=1}^{K} \beta_k^c x_{nk}^i + \sum_{e=1}^{E} \beta_e^{ci} x_{ne}^{ci} \tag{8}$$

Here, x_{ne}^{ci} represents the embedded term corresponding to the contribution of embedding base e to alternative i for latent class c. It is an element of the matrix obtained by multiplying the selectively replicated embedding input $\mathbf{e}_n$ by the class-specific embedding weight matrix $\mathbf{M}^c$ (as described in **Figure 1**). The resulting class-specific utilities V_{ni}^c across alternatives are then passed through a logit (SoftMax) function, denoted $\sigma(\cdot)$, to yield the class-specific choice probability vector $\hat{\mathbf{y}}_n^c = [\hat{y}_{n1}^c, \hat{y}_{n2}^c, \hat{y}_{n3}^c, \dots, \hat{y}_{nJ}^c]$, as shown in **Equation (9)**:

$$\hat{\mathbf{y}}_n^c = \sigma(\text{concatenate}(\mathbf{X}_n^T, R(\mathbf{e}_n) \times \mathbf{M}^c)^T \times \boldsymbol{\beta}^c) \tag{9}$$

Next, the unconditional choice probability for individual n, denoted $\hat{\mathbf{y}}_n$ (the class specifier c dropped from $\hat{\mathbf{y}}_n^c$), is obtained by weighting the class-specific choice probabilities $\hat{\mathbf{y}}_n^c$ with the corresponding class membership probability $\hat{\pi}_n^c$—computed from the DNN membership model Λ from **Equation (3)** that takes the segmentation input $\mathbf{s}_n$. This process is equivalent to **Equation (2)**, and formulated in **Equation (10)**.

$$\hat{\mathbf{y}}_n = \sum_{c=1}^{C} \hat{\pi}_n^c \hat{\mathbf{y}}_n^c, \textit{where } \hat{\pi}_n^c = \Lambda(\mathbf{s}_n)[c] \tag{10}$$

Finally, all parameters in the DCM-SEAL architecture—including the weights $\mathbf{W}$ and biases $\mathbf{b}$ of the DNN Λ, the elements of the class-specific embedding matrices $\mathbf{M}^c$'s, and the class-specific linear parameters $\boldsymbol{\beta}^c$ (including coefficients for embedded terms)—are jointly optimized through backpropagation using the AdamW optimizer. Regularization terms described in **Section 3.2** are applied (with an addition of embedding weight decay λ_e), leading to the final optimization objective shown in **Equation (11)**, where the cross-entropy loss is defined as $\mathcal{L}(\cdot) = -\mathbb{E}_n[\mathbf{y}_n \cdot \log\hat{\mathbf{y}}_n]$. The complete model architecture and computational flow are illustrated in **Figure 3**.

$$Min\ L: \mathcal{L}(\hat{\mathbf{Y}}, \mathbf{Y}; \mathbf{W}, \mathbf{b}, \boldsymbol{\beta}^c, \mathbf{M}^c) + \Omega(\mathbf{W}, \mathbf{M}^c; p_d, \lambda_s, \lambda_e, N_B) \tag{11}$$

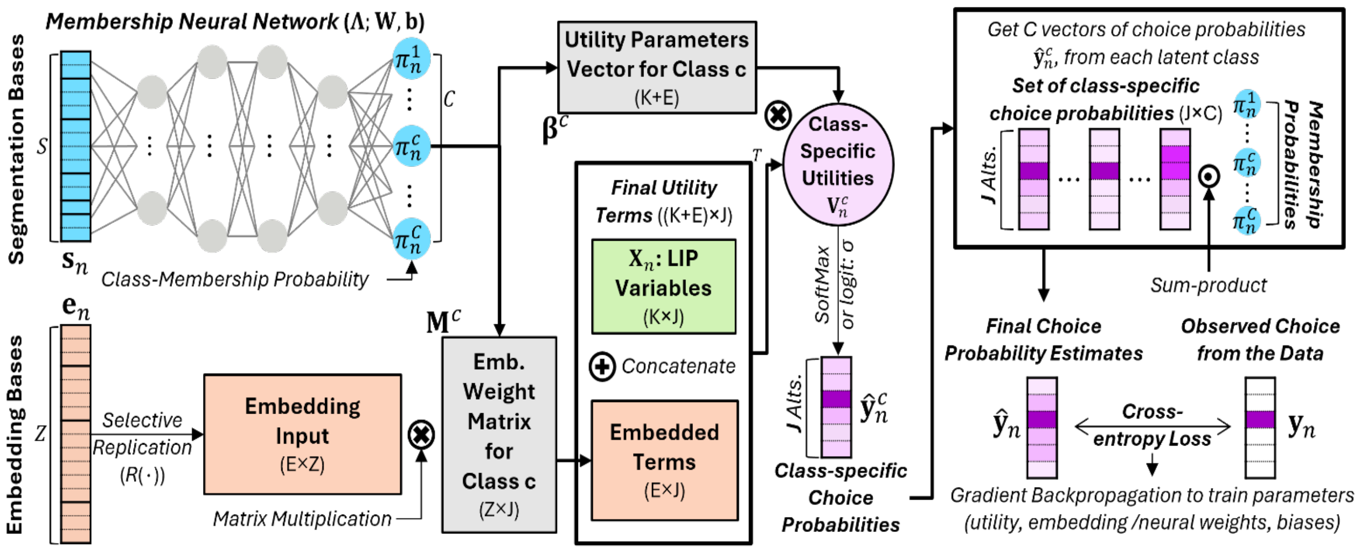

Figure 3 Overall computational flow of the DCM-SEAL architecture

3.4.3. Ensuring Practicality: Generalization and Constraints

The DCM-SEAL framework is designed as a generalized and extensible umbrella model capable of reproducing several existing discrete choice formulations as special cases under specific configurations:

- When $C = 1$ (disabling the segmentation module) and $E > 0$, this yields the **E-MNL** model (Arkoudi et al., 2023).
- When $C = 1$, $E = 0$ (no embedding module), and omitting all regularization terms, the model reduces to a **standard MNL**.
- When $C > 1$, $E = 0$, and the DNN hyperparameter $A = 0$ (no hidden layers), with no regularization, this reproduces a **conventional LCCM** with a logit-based class membership function.
- When $C > 1$, $E = 0$, and $A > 0$, this produces a DNN-membership LCCM or **NN-LCCM**, similar to Han (2019).
- Finally, when $C > 1$, $E > 0$, and $A > 0$, this constitutes the fully integrated **DCM-SEAL** framework proposed in this study.

To ensure practical applicability and numerical stability, several optional stabilization techniques are incorporated into DCM-SEAL. These are necessary because data-driven DCMs that employ DNNs inherently involve stochastic optimization, which often leads to unstable outcomes. This means that even with identical hyperparameter settings, different trials of model training may yield different parameter estimates across runs (Ali et al., 2023; Arkoudi et al., 2023; Feng et al., 2024; Han et al., 2022; Lee et al., 2018; Wang et al., 2020; Wong and Farooq, 2021). Such variability arises partly from random initialization of learnable weight/bias parameters, as well as from the stochastic nature of optimizers such as AdamW, which do not guarantee convergence to the global minimum. The multiplicative propagation of gradients can also cause *exploding* or *vanishing gradients*, occasionally amplified by perturbations at specific training steps (see Wong and Farooq (2021), for a detailed discussion).

Although various regularization techniques adopted in DCM-SEAL mitigate these issues, the resulting instability has historically motivated researchers to report average performance metrics or parameter estimates across multiple model runs (Arkoudi et al., 2023; Biswal et al., 2025; Han et al., 2022; Sifringer et al., 2020; Wang et al., 2023). This instability is further linked to interpretability challenges and to behavioral irregularities in the resulting parameters, as discussed in earlier sections.

To address these concerns, although they do not perfectly resolve the issues, DCM-SEAL incorporates several additional safeguards. First, gradient clipping and element-wise clamping are applied within each training iteration to prevent unstable updates. The global gradient norm can be set to be upper-bounded, and selected parameters (e.g., embedding weights and segmentation logits) are lower-bounded to ensure numerical stability. Second, an optional *membership entropy* term can be added to the loss function to regulate segmentation behavior. When activated, this term promotes more diffuse class memberships, helping to prevent premature convergence to a single dominant class (i.e., deterministic latent class assignment). Third, the embedding normalization scheme also differs from that of Arkoudi et al. (2023). In their E-MNL model, which employed Adam, embeddings are normalized to unit length *within each forward pass*, ensuring that gradients are constrained directly on the unit hypersphere. DCM-SEAL applies normalization *after each training epoch*, effectively projecting the updated embeddings back onto the unit sphere. This projected-gradient approach is computationally lighter and fully compatible with modern optimizers such as AdamW, offering better scalability for any of the embedding-related dimensions (E, Z, J) or the number of latent classes C (under the class-specific embedding setting) increase, while still maintaining bounded, hypersphere-constrained embeddings given a sufficient number of iterative updates.

Finally, DCM-SEAL allows optional sign constraints on specific coefficients or linear parameters (some elements of $\boldsymbol{\beta}^c$) to enhance behavioral interpretability. The parameters associated with core behavioral variables (e.g., travel time, cost) can be constrained to be non-positive, while those linked to the embedded terms are always constrained to be non-negative. These restrictions are implemented using a Softplus ($\ln(1 + \exp(x))$ function-based reparameterization, where unconstrained coefficients are internally mapped through smooth, differentiable transformations that ensure the desired sign without disrupting gradient continuity. For example, negative constraints are imposed by applying a negated Softplus transformation to a free parameter, producing a smoothly bounded negative domain. Unlike conventional DCMs that rely on non-differentiable step or piecewise transformations (e.g., triangle or threshold functions), the Softplus mapping preserves gradient smoothness and numerical stability during optimization, allowing the model to learn behaviorally consistent parameters while remaining fully compatible with backpropagation and modern optimizers within the *PyTorch* (Paszke et al., 2019) framework where DCM-SEAL is developed on.

3.5. Hyperparameter Tuning with Optuna

This subsection introduces a hyperparameter tuning framework that systematically explores the configuration space of DCM-SEAL. The procedure supports both manual and grid-search-based variable-role assignments (**Section 3.4.1**) by efficiently identifying optimal hyperparameter combinations according to held-out evaluation performance and to retrieve behavioral plausibility models. Hyperparameter tuning is a critical step in ML model development, as these parameters are not optimized within the model itself, but instead serve as externally specified settings that strongly influence training dynamics and predictive performance.

The importance of systematic tuning has been emphasized across multiple data-driven DCM studies: Ali et al. (2023) tested 100 distinct hyperparameter configurations to identify the best-performing model; Han (2019) examined multiple settings of four key hyperparameters while monitoring performance on the hold-out *validation dataset*; and Salvadé and Hillel (2025) implemented an automated tuning software Hyperopt (Bergstra et al., 2015) to optimize model behavior. As noted by Baek and Khani (2026), substantial gains in model accuracy can be achieved through careful design of the search space and tuning strategy, underscoring that systematic hyperparameter optimization is not optional but essential for achieving reliable and reproducible performance in complex, data-driven DCMs.

In DCM-SEAL, the *Optuna* framework (Akiba et al., 2019) is employed to automatically explore the hyperparameter space and identify configurations that jointly balance predictive accuracy, numerical stability, and behavioral interpretability. Optuna is favored for its rapid convergence, customizable objective definitions, and integrated early-stopping (pruning) functionality. Its sampling algorithms, including the Tree-Structured Parzen Estimator (TPE) and the Covariance Matrix Adaptation Evolution Strategy (CMA-ES), can operate independently or in hybrid mode, allowing both exploratory and exploitative search behaviors. The combination of those techniques and the pruning implemented through the Asynchronous Successive Halving Algorithm (ASHA) enables scalable, parallelizable optimization that maintains statistical rigor while reducing computational overhead.

In practice, each Optuna trial corresponds to a complete model training run in which a candidate set of hyperparameters is sampled, evaluated, and updated based on feedback from an evaluation metric or objective defined by the user. For DCM-SEAL, the parameters tuned in this process include both those governing the DNN segmentation module and the global training procedure (the seven parameters introduced in **Section 3.2**), along with an embedding-specific extension of additional weight-decay coefficient λ_e, and a binary flag indicating whether the model should employ class-specific or shared embedding matrices (i.e., either to define a global $\mathbf{M}$ shared across all classes, versus class-specific $\mathbf{M}^c$'s). To avoid computationally heavy configurations, we also introduce pseudo-hyperparameters N_C (the number of computations) and N_U (the number of updates per epoch) drawn under Optuna's automated algorithm, which is used to derive the number of epochs (N_E) and the number of mini-batches (N_B) via simple capped divisions. The complete search space for all tunable parameters is summarized in **Table 1**.

Table 1 Hyperparameter search ranges and sampling methods used in the Optuna-based tuning of DCM-SEAL

Category	Hyperparameters drawn from Optuna and their primary role	Range or category of sampling	Space drawn from
Global	N_C: The number of computations [a]	[300, 600, 900, 1200, 1500]	Categorical
	N_U: The number of mini-batch updates per epoch (1: full-batch) [b]	[1, 10, 20, 40]	Categorical
	γ: Gradient descent learning rate	From 0.001 to 0.1	Numeric (log-scale)
Class Membership DNN	A: The number of hidden layers	From 2 to 4	Integer
	N_a ($a \in \{1,2,3, \dots A\}$): the number of nodes of the a-th hidden layer	[64, 128, 256]	Categorical
	p_d: Dropout probability for DNN weights	From 0 to 0.4	Numeric
	λ_s: Weight decay for DNN weights	From 0.0001 to 0.01	Numeric (log-scale)
Embedding	λ_e: Weight decay for embedding weights (elements of $\mathbf{M}$ or $\mathbf{M}^c$'s)	From 0.0001 to 0.01	Numeric (log-scale)
	Class-specific embedding flag: whether to estimate $\mathbf{M}$ or $\mathbf{M}^c$'s	["shared" or $\mathbf{M}$, "class specific" or $\mathbf{M}^c$s]	Categorical
Fixed Settings	Non-positivity constraints on time or cost coefficients (i.e., selected β_k^c's), non-negativity constraints for embedding coefficients β_e^{ci}'s via Softplus mapping, and L2 normalization for all embedding matrices' rows after each training epoch		

[a] This is a pseudo-hyperparameter only used to derive the actual hyperparameter N_E, the number of epochs, via $\max(N_C/N_U, 500)$.
[b] This is a pseudo-hyperparameter used to derive the above N_E and another hyperparameter N_B, the mini-batch sample size, via $\min(\max(128, N/N_U))$.

Automatic hyperparameter tuning is particularly crucial for DCM-SEAL because the model simultaneously handles multiple interacting components and varying variable-role assignments, which collectively require fast hyperparameter convergence for each scenario. Combining the grid-search role-exploration strategy (**Section 3.4.1**) and the required experiments with different numbers of latent classes (iterating C values to vary from 2 to 4; see the last paragraph of **Section 2**), each experiment instance of SB/EB split and varying C constitutes a distinct *decision scenario*. For every scenario, Optuna conducts multiple (50-200) trials that sample hyperparameters from the predefined ranges listed in **Table 1**, drawing values based on prior results. This multi-layered search over decision scenarios and hyperparameters ensures that the final model configuration reflects both the structural and numerical optima, rather than relying on ad hoc parameter choices.

Finally, the objective of each Optuna trial is set to minimize the training held-out *test dataset*'s log-likelihood or $\sum_n(\mathbf{y}_n \cdot \log\hat{\mathbf{y}}_n)$, which contrasts with standard ML classification settings that often rely on precision, recall, or F1 scores. This is because DCM-SEAL's ultimate goal is not mere categorical prediction but behaviorally consistent and transferable model estimation, thus the log-likelihood and its related metric like McFadden's ρ^2 better reflects goodness-of-fit and the interpretability of the resulting parameters (Ali et al., 2023).

4. DCM-SEAL Experiment on Synthetic Dataset

This section evaluates the proposed DCM-SEAL framework by applying it to a synthesized dataset to demonstrate its ability to recover known behavioral parameters. The synthetic dataset is generated via Monte Carlo simulation, in which ground-truth parameters, class memberships, and choice outcomes are fully known. Class memberships are generated semi-deterministically from auxiliary variables, and individual choices are simulated according to intertwined class-specific utility functions that reflect a plausible transit mode choice context. Specifically, endogeneity is deliberately incorporated in this design, making it difficult for conventional MNL or even standard LCCM formulations to recover the true membership structure and behavioral parameters. This dataset, therefore, provides a rigorous testbed for demonstrating DCM-SEAL's capacity to recover nonlinear, heterogeneous, and endogenous behavioral patterns.

4.1. Synthetic Data Design and Generation

Our synthetic dataset was constructed to evaluate the behavioral interpretability and parameter recovery accuracy of DCM-SEAL. The dataset provides a controlled environment in which the true parameter values and class memberships are known, allowing systematic comparison across modeling frameworks. The binary synthetic design is intentionally simple because its primary purpose is parameter recovery under controlled endogeneity and latent heterogeneity, not to maximize the empirical advantage of the adopted ML components in a large multi-alternative setting.

The data-generation process explicitly incorporates endogeneity, thereby emulating challenges commonly encountered in real-world mode-choice modeling. Each synthetic observation represents an individual making a binary travel-mode choice between the urban *bus* and *train*. Latent-class memberships are assigned semi-deterministically based on auxiliary demographic and contextual variables, which are themselves drawn from prespecified marginal distributions, except for one, which is drawn from a correlated distribution. Class-specific utility functions are hand-crafted to reflect realistic behavioral differences, such as varying sensitivity to the Core variables of the typical mode-choice problem and a latent variable that both affects class membership and utility in a nonlinear, interactive manner. Full numerical details of the synthesis process are provided in **Appendix A**, while this section outlines its conceptual design and focuses on the derivation of ground truth choice utility.

The synthesis begins by drawing 12,500 random values from an exponential distribution, which serve as an implicit variable *travel distance* used to generate three core travel attributes. Based on the distance, in-vehicle time (*IVT*: x_1) for bus and train are generated from linear relationships with additional random "jitter" terms specific to each mode. The incremental rate of IVT growth with distance is higher for buses than for trains, ensuring that bus IVT is typically longer. However, because the bus-specific random term has higher variance, some realizations yield shorter bus IVT than train IVT for the same distance. Out-of-vehicle time (*OVT*: x_2), the sum of waiting and walking time, is generated from simple but distinct distributions for the two modes, where train's is generally longer than that of bus. The final core variable, *fare* (x_3), is also distance-based: train has a higher base fare but a lower distance-incremental rate, again with added randomness to introduce sample diversity.

Four sets of individual-specific categorical variables are then generated for each observation using discrete probability mass functions: favorite colors (o_1;4 levels), race (s_1; 3 levels), gender (s_2; 3 levels), and trip purpose (e_1; 3 levels). As the notations suggest, variable o_1 is intentionally made irrelevant to utility and included as a "redundant" variable. Variables s_1 and s_2 influence the latent class assignment, while e_1 is a non-core categorical variable that affects utility but is not used for direct marginal rate-of-substitution (MRS) estimation. A fifth categorical variable, income (e_2; 3 levels), is probabilistically generated from a constant probability matrix linking race (s_1) and income levels, imposing an interaction between variables reflecting a society in which certain racial groups are more likely to belong to higher-income categories.

To simulate mode choice, a naïve utility function is first formulated for the binary train versus bus decision, as shown in **Equation (12)** (for simplicity, the respondent index n is omitted where not needed). The bus is treated as the reference alternative, so the alternative-specific constant (ASC) only applies to and is defined for the train. Using the indicator function $\mathbb{1}_{\{\}}$, which returns 1 if the condition in brackets is true and 0 otherwise, both modes' utilities can be expressed in a single formulation:

$$U = \mathbb{1}_{\{train\}} ASC + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \epsilon = \mathbb{1}_{\{train\}} ASC + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_q q + \varepsilon \tag{12}$$

As shown on the right-hand side of **Equation (12)**, an unobserved latent variable, *Safety concern* (q), is introduced to represent individuals' perceived risk or discomfort during their transit trip. We assumed for some travelers, q increases with IVT or x_1 (e.g., claustrophobic discomfort), while for others it increases with OVT or x_2 (e.g., safety concerns in high-crime neighborhoods during walking or waiting). This structure induces endogeneity, since the mis-specified error term $\epsilon = \beta_q q + \varepsilon$ is actually correlated with x_1 and x_2 through q, where ε is the true independent error term. Such endogeneity can be partially mitigated via a latent-class formulation that captures systematic taste heterogeneity (Kim and Mokhtarian, 2018); readers are referred to Guevara (2015) for a detailed treatment of endogeneity in DCMs.

To design heterogeneous class-specific utilities, latent variable q is first expressed as a linear combination of latent parameters $\alpha_1, \alpha_2, \alpha_3, \alpha_4$ as shown in **Equation (13)**, where superscripts on categorical variables indicate their active levels (binary 0-1 indicators). For example, individuals with $s_1^2 = 1$ (of a specific gender) are assumed to perceive train travel as safer ($\alpha_1 < 0$) and to have increased concern with longer IVT ($\alpha_2 > 0$) for both modes. Similarly, individuals with $s_1^1 = 1$ and $s_2^1 = 0$ (a combination of gender and race) are assumed to experience growing safety concerns with OVT ($\alpha_3 > 0$), an effect partially offset for those with specific income category ($e_2^3 = 1$) by parameter α_4.

$$q = s_1^2\left(\alpha_1 \mathbb{1}_{\{train\}} + \alpha_2 x_1\right) + s_1^1(1 - s_2^1)(\alpha_3 x_2 + \alpha_4 e_2^3) + \delta \tag{13}$$

The mean-zero, normally distributed error term δ adds randomness to class assignment, producing semi-deterministic membership. Each observation is probabilistically assigned to a behaviorally known class based on s_1 and s_2, using a logit-type nonlinear interaction formula that defines the average nominal or target class membership probability $\overline{\mathrm{P}}(n^c)$, shown in **Equation (14)**, where the actual latent class membership assignment is implemented through the simulation based on these probabilities.

$$\overline{\mathrm{P}}(n^1) = 0.95 s_{1,n}^2;\ \overline{\mathrm{P}}(n^2) = 0.95 s_{1,n}^1(1 - s_{2,n}^1);\ \overline{\mathrm{P}}(n^3) = 0.95(1 - s_{1,n}^2)(1 - s_{1,n}^1(1 - s_{2,n}^1)) \tag{14}$$

Finally, categorical variables e_1 and e_2 (intended to be included in EB) are incorporated into the utility formulation for the non-reference alternative (train), with each level assigned its own parameter with superscript (β_k^l: k denoting variables and l unique levels). The resulting class-specific utility functions U^c for $c = 1,2,3$ are expressed in **Equations (15)-(17)**, where the interactions of safety concern q with other variables are linearly decomposed. Therefore, we can expect that a correctly specified latent class model should be able to recover these relationships, for example, by estimating $\hat{\beta}_1^{c=1} \approx \beta_1 + \alpha_2$ and $\hat{\beta}_2^{c=2} \approx \beta_2 + \alpha_3$, while $\hat{\beta}_3^{c=3} \approx \beta_3$. Finally, the random error term ε is drawn from a normal distribution $N(0, (1.5\beta_1)^2)$, and the choice outcome $\mathbf{y}_n$ is determined by selecting the alternative with the higher simulated utility.

$$U^1 = 1_{\{train\}}(ASC + \alpha_1 + \Sigma_{k=1}^2 \Sigma_{l=1}^3 \{\beta_k^l e_k^l\}) + (\beta_1 + \alpha_2)x_1 + \beta_2 x_2 + \beta_3 x_3 + \varepsilon \tag{15}$$

$$U^2 = 1_{\{train\}}(ASC + \Sigma_{l=1}^3 \{\beta_1^l e_1^l\} + \beta_2^1 e_2^1 + \beta_2^2 e_2^2 + (\beta_2^3 - \alpha_4)e_2^3) + \beta_1 x_1 + (\beta_2 + \alpha_3)x_2 + \beta_3 x_3 + \varepsilon \tag{16}$$

$$U^3 = 1_{\{train\}}(ASC + \Sigma_{k=1}^2 \Sigma_{l=1}^3 \{\beta_k^l e_k^l\}) + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \varepsilon \tag{17}$$

The ground-truth marginal rates of substitution (MRS) for each parameter, normalized by the IVT coefficient (β_1), are summarized in **Table 2**. These values serve as reference targets for evaluating DCM-SEAL's parameter recovery accuracy in **Section 4.3**.

Table 2 Ground-truth marginal rates of substitution (MRS) for parameters in the synthetic dataset, relative to in-vehicle time (N=12,500)

Name	Notation	Description	Assumed, True MRS
Alternative specific constant for the mode *Train*	ASC	Alternative specific constant (reference mode: bus)	-6
x_1: In-vehicle time (IVT)	β_1	Associated with an LIP variable	1
x_2: Out-of-vehicle time (OVT)	β_2	Associated with an LIP variable	1.5
x_3: Fare	β_3	Associated with an LIP variable	0.5
e_1: Trip Purpose (3 levels)	$\beta_1^1, \beta_1^2, \beta_1^3$	Associated with an embedding base	-10, -2, 2
e_2: Income (3 levels)	$\beta_2^1, \beta_2^2, \beta_2^3$	Associated with an embedding base	-1, -2, -4
Safety perception for train (for Class 1)	α_1	Primarily interacting with ASC, s_1	-1.5
Safety concern for incremental IVT (for Class 1)	α_2	Primarily interacting with x_1, s_1	0.25
Safety concern for incremental OVT (for Class 2)	α_3	Primarily interacting with x_2, s_1, s_2	1
Safety concern offset (for selected individuals in Class 2)	α_4	Applied to Class 2 people with $e_2^3 = 1$	5

* Note: ASC and parameters for embedding bases (β_1^l's and β_2^1's) are included only in the train's utility function for simplicity (corresponding effect for bus mode can be treated as 0, as only the difference in utility matters); no parameters are defined for any levels of the two segmentation bases (s_1: gender, s_2: race) by definition: conceptually, they are only affecting the latent variable (safety concern) interactions, and technically, only affecting the latent class assignment.

4.2. Synthetic Data Experiment Setup

Using the synthetic dataset, we estimated six types of models designed to benchmark the performance and behavioral interpretability of DCM-SEAL against conventional DCMs and recent data-driven formulations, where for all models, x_1, x_2, and x_3 are treated as core-LIP variables:

- $\mathcal{M}_1$: A **baseline MNL** model including only the three core-LIP variables (x_1, x_2, and x_3).
- $\mathcal{M}_2$: A **full MNL** model incorporating one-hot encoded versions of all five categorical variables (s_1, s_2, e_1, e_2, and o_1) with 16 total unique levels, each entering linearly in the utility function.
- $\mathcal{M}_3$: A replicated **conventional LCCM** with three latent classes, using s_1, s_2 and o_1 as SB via a shallow ANN (equivalent to a standard logit membership function), while e_1 and e_2 enter linearly in the utility after one-hot encoding.
- $\mathcal{M}_4$: The **E-MNL** model, which assigns all five categorical variables as EB.
- $\mathcal{M}_5$: A three-class neural-membership LCCM (**NN-LCCM**), assigning all five categorical variables to the SB role.
- $\mathcal{M}_6$: A three-class **DCM-SEAL** model, with s_1, s_2 and o_1 specified as SB and e_1, e_2 as EB.

The synthetic dataset was divided into training (80%) and test (20%) subsets. Each model was trained on the training set, and predictive performance on the test set was evaluated using McFadden's pseudo R^2 or ρ^2 statistic, computed as $\rho^2 = 1 - LL(\beta)/LL(0)$[3] (unless specified otherwise, this ρ^2 statistic from the test data is hereinafter referred to as *accuracy* or *model performance*). For the conventional MNL models ($\mathcal{M}_1$ and $\mathcal{M}_2$), estimation was conducted using the *R* package *mlogit* (Croissant, 2020). The remaining four models, which cannot be directly specified in mlogit, were implemented and optimized within our DCM-SEAL PyTorch framework. There, to replicate the conventional LCCM structure for $\mathcal{M}_3$, all regularization parameters were set to zero, and full-batch gradient updates were enforced. Detailed experiment settings are described in the following subsections.

4.2.1. Parameter Identification and Reporting Metrics

Parameter identifiability is a critical consideration when comparing models across different estimation frameworks. In conventional DCMs (e.g., models trained on mlogit), identification is ensured by normalizing the variance of the Gumbel-distributed error term to $\pi^2/6$ and by excluding perfectly collinear variables, thereby fixing the overall parameter scale and avoiding singularity in the information matrix.

In contrast, DCM-SEAL models trainable on AI-development frameworks like PyTorch do not impose these identification rules by default. The model can accommodate linearly correlated inputs, and the scale of estimated coefficients may vary across specifications due to differing loss normalizations. Therefore, for all models, we report results in terms of marginal rates of substitution (MRS), which normalize each parameter by the magnitude of its corresponding class-specific time coefficient ($\hat{\beta}_{x_1}$). This rescaling facilitates direct behavioral comparison across models.

4.2.2. Evaluating Latent-Class Recovery

In addition to evaluating the models' choice probability prediction accuracy via $LL(\beta)$ and ρ^2, for the latent class models ($\mathcal{M}_3$, $\mathcal{M}_5$, and $\mathcal{M}_6$), we computed the Mean Relative Absolute Error (MRAE) of the MRS values. These values are obtained by comparing the model-estimated behavioral parameters with the ground-truth MRS known from the data synthesis. This measure captures the relative deviation or behavioral discrepancy between the ground truth and the recoveries from each model. Since the latent parameters (α's) in the data-generation process cannot be fully estimated in linear-in-parameter utility formulation unless explicitly specified (which is practically impossible in most cases, as they are unknown), the comparison was made against the "total impact" coefficients defined in **Equations (15)-(17)**. For example, the target MRS for Class 2's out-of-vehicle time (x_2) is $\beta_2 + \alpha_3$ =2.5, rather than the base theoretical coefficient $\beta_2 = 1.5$.

In addition, because $\mathcal{M}_5$ does not have the parameters estimated for e_1 and e_2 by the model design, we computed two MRAE measures: 1) Core-MRAE, based on class-specific ASCs and the associated MRS values (of the above "total impact") with x_2 and x_3; and 2) All-MRAE, based on all estimable MRS values (27 total). Finally, the ordering of model-defined latent classes is inherently arbitrary, meaning class labels may not correspond to the true latent classes in the synthetic data. To align and label them, we applied the Hungarian algorithm (Kuhn, 1955), only on the three core parameters (ASC, x_2, and x_3) pair for each class, minimizing the total MRAE between each estimated and ground-truth MRS vector.

4.2.3. Model Fitting and Averaging

Following best practices in prior literature, we evaluated models trained from the DCM-SEAL framework ($\mathcal{M}_3$ to $\mathcal{M}_6$) using average estimates from multiple runs rather than single-run outcomes, due to the stochastic nature of neural optimization. For each model, we first identified the best-performing hyperparameter combination (with the highest test ρ^2) among 200 Optuna trials and fixed this combination for repeated training.

We then repeatedly trained each model under the pinned configuration (listed in **Appendix B**) and discarded results with test ρ^2 values below the 95th percentile observed during the 200-trial hyperparameter tuning. This is to follow the methodological guidelines established by Baek and Khani (2026), stating that reporting the raw, unfiltered distribution obtained during hyperparameter tuning trials is uninformative because optimization algorithms like Optuna rely on directed search heuristics rather than random draws. The resulting search history inherently contains highly suboptimal exploratory configurations that are never intended for physical deployment. In other words, discarding trials below the 95th percentile filters out this stochastic optimization noise, ensuring that the final baseline comparisons reflect a stable, fully converged model. Consequently, this thresholding strategy is not a source of selection bias, but a standard regularization mechanism necessary to isolate the true performance ceiling of the ML-included modeling architecture.

This process continued until 30 stable runs were obtained for each model. Across the ML-integrated models (from $\mathcal{M}_4$ to $\mathcal{M}_6$), a total of 283, 368, and 244 training runs were required, respectively, to obtain 30 stable results meeting the 95th-percentile test ρ^2 threshold. In other words, more than 10% of model runs using the tuned or optimized hyperparameter settings achieved accuracy exceeding the top 5% results from the global hyperparameter space, underscoring the critical role of systematic tuning in achieving reliable, high-performing model outcomes.

4.3. Model Results from Synthetic Data

Table 3 summarizes the model comparison results and evaluation metrics. Models $\mathcal{M}_3$ to $\mathcal{M}_6$ report both the 30-run average and best-performing results, while $\mathcal{M}_1$, $\mathcal{M}_2$, and $\mathcal{M}_4$ present class-agnostic average-behavior outcomes by their design. Variables s_1, s_2, and o_1, which were not designed to enter the utility, are shown in gray font. As discussed above, all reported behavioral measures are expressed as MRS values rather than raw coefficients.

The comparison of the best-performing runs across all models clearly demonstrates the incremental benefits of incorporating richer behavioral structures and appropriate variable-role assignments. Relative to the baseline $\mathcal{M}_1$ (Core MNL), which achieved the lowest test $\rho^2 = 0.151$, the $\mathcal{M}_2$ (Full MNL) model substantially improved fit ($\rho^2 = 0.546$) by including additional categorical variables. However, as both MNL models rely on a single linear-in-parameter utility formulation, they cannot capture the underlying class-specific taste heterogeneity generated from the intended endogeneity and non-linear interactions inherent in the dataset. This limitation becomes more evident when comparing $\mathcal{M}_2$ and $\mathcal{M}_4$ (E-MNL): although $\mathcal{M}_4$ integrates embeddings, its performance ($\rho^2 = 0.548$) remains similar to $\mathcal{M}_2$ because the synthetic dataset's categorical variables have low cardinality (maximum of four levels) and minimal sparsity, offering limited benefit from embedding representations except for its ability to let modelers compactly examine how each levels are related to which alternative.

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[3] For conventional discrete choice models estimated through analytical software, the baseline log-likelihood $LL(0)$ is typically computed using a constants-only model, where the observed shares of alternatives in the dataset define the choice probabilities. However, in machine learning-based approaches where the dataset is split into training and test subsets, using alternative-specific constants estimated on each subset can introduce reporting bias. To avoid this, we adopt a dataset-independent baseline by assuming uniform choice probabilities $LL(0) = N\log(1/J)$) which depends only on the number of observations and available alternatives, both fixed given a training-testing split ratio. In addition, while an adjusted goodness-of-fit measure $\bar{\rho}^2$ that penalizes model complexity is often reported to encourage parsimony, we rely on regularization within the ML-integrated framework to mitigate redundancy. Therefore, we report the unadjusted ρ^2 alongside the model's final log-likelihood or $LL(\beta)$ for complementary evaluation.

Table 3 Model estimation results on the synthetic dataset: comparison of parameter recovery and model fit across six specifications

Results	Ground Truth MRS [a]	$\mathcal{M}_1$: Core MNL [b]	$\mathcal{M}_2$: Full MNL [b]	$\mathcal{M}_3$: LCCM	$\mathcal{M}_4$: E-MNL	$\mathcal{M}_5$: NN-LCCM	$\mathcal{M}_6$: DCM-SEAL
Overall Model Statistics on the Test Dataset (values in parentheses: standard deviation)							
ρ^2 (average) [c]		-	-	0.538 (0.0138)	0.547 (0.0003)	0.560 (0.0067)	**0.591** (0.0021)
ρ^2 (maximum)		0.151	0.546	0.570	0.548	0.573	**0.596** [e]
Log-likelihood (average) [c]		-	-	-801 (23.95)	-785 (0.44)	-763 (11.63)	**-709** (3.7)
Log-likelihood (maximum)		-1472	-787	-745	-783	-740	**-700** [e]
Core-MRAE [d] (30-run average)		-	-	1.30 (0.504)	-	**0.76** (0.41)	1.00 (0.726)
Core-MRAE [d] (minimum)		-	-	0.65	-	0.39	**0.31** [e]
All-MRAE [d] (30-run average)		-	-	1.41 (0.362)	-	-	**1.37** (0.478)
All-MRAE [d] (minimum)		-	-	0.76	-	-	**0.51**
Latent Class 1 or Single-class Average: Parameter Estimates' Mean Marginal Rate of Substitution for Associated Variable							
ASC	-5.0	-12.05	-4.31	-1.18	-3.27	-4.34	-2.96
x_1 (reference)	1.0	1.00	1.00	1.00	1.00	1.00	1.00
x_2	1.2	1.34	1.42	0.83	1.42	0.98	0.91
x_3	0.4	0.83	0.64	0.72	0.60	0.57	0.50
e_{11}	-8.0	-	-11.41	-3.43	-6.13	-	-5.37
e_{12}	-1.6	-	-4.03	0.90	1.22	-	-0.15
e_{13}	1.6	-	-	2.97	5.39	-	2.81
e_{21}	-0.8	-	2.51	-0.83	-2.39	-	-1.52
e_{22}	-1.6	-	1.39	-2.04	-3.35	-	-2.33
e_{23}	-3.2	-	-	-2.36	-4.78	-	-3.28
s_{11}	-	-	1.76	-	1.12	-	-
s_{12}	-	-	-1.67	-	-2.38	-	-
s_{13}	-	-	-	-	-0.70	-	-
s_{21}	-	-	-3.28	-	-3.41	-	-
s_{22}	-	-	-0.07	-	-0.11	-	-
s_{23}	-	-	-	-	-0.07	-	-
o_{11}	-	-	-0.11	-	-0.70	-	-
o_{12}	-	-	-0.17	-	-0.74	-	-
o_{13}	-	-	-0.26	-	-0.85	-	-
o_{13}	-	-	-	-	-0.62	-	-
Latent Class 2: Parameter Estimates' Mean Marginal Rate of Substitution for Associated Variable							
ASC	-6.0	-	-	-2.62	-	-5.60	-3.87
x_1 (reference)	1.0	-	-	1.00	-	1.00	1.00
x_2	2.5	-	-	2.32	-	2.24	2.99
x_3	0.5	-	-	2.33	-	0.76	2.07
e_{11}	-10.0	-	-	-2.31	-	-	-8.42
e_{12}	-2.0	-	-	0.42	-	-	-0.03
e_{13}	2.0	-	-	2.56	-	-	4.71
e_{21}	-1.0	-	-	-0.71	-	-	-3.09
e_{22}	-2.0	-	-	-0.97	-	-	-4.32
e_{23}	1.0	-	-	-1.76	-	-	-5.81
Latent Class 3: Parameter Estimates' Mean Marginal Rate of Substitution for Associated Variable							
ASC	-6.0	-	-	-1.75	-	-4.63	-2.98
x_1 (reference)	1.0	-	-	1.00	-	1.00	1.00
x_2	1.5	-	-	1.02	-	1.00	1.97
x_3	0.5	-	-	1.64	-	0.79	1.13
e_{11}	-10.0	-	-	-3.31	-	-	-6.25
e_{12}	-2.0	-	-	1.39	-	-	-0.04
e_{13}	2.0	-	-	2.02	-	-	3.46
e_{21}	-1.0	-	-	0.05	-	-	-2.49
e_{22}	-2.0	-	-	-1.13	-	-	-3.39
e_{23}	-4.0	-	-	-2.14	-	-	-4.46

[a] The "Ground Truth MRS" column represents the total impact for each latent class, derived by substituting the canonical parameters from **Table 2** into Equations **(15)**-**(17)**.

[b] The two models were estimated on the R package *mlogit*

[c] Note that the statistics are based only on the first 30 trials that met each model's performance threshold, so interpretation should be made with caution.

[d] MRAE denotes the Mean Relative Absolute Error of the marginal rates of substitution (MRS): Core-MRAE is computed from the 9 core parameters common to all latent-class models, while All-MRAE includes all 27 estimable parameters (excluding the normalized reference term x_1).

[e] The model run with the maximum ρ^2 corresponds to the same run that yielded the minimum MRAE, indicating that the best-fitting model also achieved the most accurate parameter recovery.

Specifically, the original E-MNL paper states in low-cardinality environments where categorical variables possess fewer than five unique levels and are well-represented across alternatives, conventional dummy encoding does not trigger overfitting or parameter explosion. Under these conditions, a linear-in-parameter model with dummy variables can perform equally well (Arkoudi et al., 2023). On the other hand, $\mathcal{M}_4$ exhibited one of the smallest standard deviations of test ρ^2across 30 runs, indicating greater numerical stability and faster convergence.

Introducing latent classes remarkably improved model performance. The replicated LCCM ($\mathcal{M}_3$) achieved $\rho^2 = 0.570$, already outperforming all single-class models by capturing discrete taste heterogeneity through segmentation. The data-driven NN-LCCM ($\mathcal{M}_5$) slightly improved model fit ($\rho^2 = 0.572$) by learning non-linear interactions in class membership via the neural network, partially due to the limited, non-complex interactions generated in the data synthesis that may not require deep structure. Finally, the proposed DCM-SEAL ($\mathcal{M}_6$) achieved the highest overall test performance ($\rho^2 = 0.596$), significantly outperforming any other best runs from other modeling specifications.

Regarding the minimum Core-MRAE obtained from each model specification, the lowest value of 0.31 was found for $\mathcal{M}_6$, followed by $\mathcal{M}_5$ (0.39), both substantially lower than the baseline $\mathcal{M}_3$'s 0.65. For the All-MRAE statistics, the difference between $\mathcal{M}_6$ (0.51) and $\mathcal{M}_3$ (0.76) is relatively smaller, but DCM-SEAL still outperforms the conventional model. These results confirm that integrating ML components into DCMs substantially enhances their ability to recover the true behavioral structure implicit in the data. Moreover, we have observed that compared to $\mathcal{M}_5$, the reallocation of selected variables from SB to EB in $\mathcal{M}_6$ not only improved model fit but also enhanced behavior recovery accuracy. This underscores that embedding serves best as a complementary mechanism for context-dependent variables rather than class-defining ones (i.e., each variable may have its "recommended" EB/SB membership). This point is further discussed in the SwissMetro analysis in **Section 5**.

Taken together, these findings highlight two main insights. First, endogeneity and taste heterogeneity cannot be adequately addressed by simply expanding linear utility specifications; models that incorporate latent class segmentation and nonlinear mechanisms yield more behaviorally coherent and accurate predictions. Second, the balance of variable-role assignments—particularly the division of non-core variables between SB and EB—has a pronounced influence on both model stability and interpretability. Among the ML-integrated models ($\mathcal{M}_4$ to $\mathcal{M}_6$), assigning all variables to SB to make it an NN-LCCM, tended to reduce model predictive performance, whereas DCM-SEAL's hybrid configuration achieved the most accurate and moderately interpretable performance.

Similar conclusions emerge from the 30-run statistics, though these results should be interpreted with caution. This is because in ML practice, it is common to retain only the best-performing or most practically useful models, discarding less effective trials. As noted by Hastie et al. (2009) and Bishop (2006), model selection based on predictive performance is a standard component of data-driven model development. Accordingly, the comparison of the best-performing models presented above can provide a more meaningful evaluation than average statistics alone. The 30-run averages, therefore, represent outcomes of a selection process that captures the expected number of runs required to obtain a "good" and stable model and the empirical range of their values, rather than fully reproducible results.

Nevertheless, the average test ρ^2 values show that $\mathcal{M}_5$ and $\mathcal{M}_6$ clearly outperform $\mathcal{M}_4$, while the standard deviation of ρ^2 was the smallest in $\mathcal{M}_4$, followed by $\mathcal{M}_6$ and the largest in $\mathcal{M}_5$. This pattern suggests a trade-off between flexibility and numerical stability: as the number of neural network parameters increases exponentially with higher input dimensionality, the optimization process becomes more complex and prone to variance across runs. Finally, regarding the 30-run average MRAE results, both $\mathcal{M}_5$ and $\mathcal{M}_6$ outperform $\mathcal{M}_3$ in terms of core parameters, and $\mathcal{M}_6$'s All-MRAE remains lower than that of $\mathcal{M}_3$, aligning with the above findings from the comparison of the best-performing models.

Given the synthetic nature of the dataset (i.e., the data and results are not implying the actual stated/revealed preference and thus do not show meaningful travel behavior) and the MRS values are derived ones from a division between estimates, we refrain from interpreting individual class-specific MRS estimates in **Table 3** and doing statistical tests, unlike the subsequent section's experiment with the real-world data.

5. DCM-SEAL Experiment on SwissMetro for Benchmarking and Variable Role Discussion

This section presents an evaluation of the proposed DCM-SEAL framework by applying it to SwissMetro dataset (Bierlaire et al., 2001). The dataset serves as a real-world benchmark, enabling comparisons with established studies and the derivation of behavioral insights. After preprocessing (e.g., removal of incomplete choice sets), the model is ready to be trained and evaluated under conditions comparable to those in the prior literature. The use of SwissMetro is intended to showcase how DCM-SEAL can be applied to existing choice data and benchmark its predictive and behavioral performance against other data-driven DCMs. In addition, we explore the impacts of variable role selection, or "split," to inform the future modeling strategy.

5.1. Dataset Overview

The SwissMetro dataset is a publicly available, widely used benchmark in transportation research, serving a critical role in evaluating various discrete-choice studies. The dataset originated from a two-stage revealed-preference and stated-preference survey designed to assess the potential demand for and acceptance of a proposed maglev train, SwissMetro (SM), in Switzerland's regional travel market. In the surveyed choice task, respondents selected one of three alternatives: Train, SM, or Car. For each alternative, key explanatory variables included door-to-door travel time, fare/cost, and service headway.

The dataset has also become a cornerstone for evaluating the predictive performance and behavioral consistency of emerging modeling frameworks, particularly neural-network-integrated data-driven DCMs. While the original survey produced 10,728 observations, studies have adopted different cleaning and partitioning procedures to ensure comparability and computational tractability. For instance, Sifringer et al. (2020) and Arkoudi et al. (2023) relied on a reduced subset containing only complete observations with full variable coverage. Regardless of the specific preprocessing scheme, the dataset's primary value lies in benchmarking models using metrics such as log-likelihood (LL) or cross-entropy loss (CEL) on held-out test data, as well as traditional behavioral indicators like VoT.

To assess the stochastic variability inherent in DNN-based models, prior SwissMetro studies have commonly performed multiple training runs with different random initializations. For example, Han et al. (2022) repeated training up to 100 times, while Arkoudi et al. (2023) averaged reported outcomes over 30 runs. Similarly, Salvadé and Hillel (2025) repeated training 20 times in their RUMBoost model. Other recent studies, including MNL-ARD (Rodrigues et al., 2022) and DCM-LN (Kim and Bansal, 2024), have also employed the SwissMetro data to test interpretability, underscoring the dataset's importance for verifying robustness and behavioral realism across data-driven DCMs.

For the application of DCM-SEAL, we adopted the preprocessing procedure used by Sifringer et al. (2020) and Arkoudi et al. (2023). Among the 10,728 original records, 9,036 observations containing complete information were retained. The filtered dataset was then randomly divided into 7,234 training samples (80%) and 1,802 test samples (20%). Three continuous travel-time- and cost-related variables, namely CO (cost), TT (travel time), and HE (headway), were designated as core and thus assigned to the LIP role. These three core variables ($K_0 = 3$) along with two ASCs yield a total of 5 parameters, creating the input LIP dimension $K = 4$ (3+ASC column) by $J = 3$ alternatives. The remaining 12 categorical or ordinal variables were first classified as non-core and assigned to either the Segmentation Bases (SB) or Embedding Bases (EB), depending on the selected variable-role (split) strategy and each experiment configuration.

Under the manual split strategy, considering the downstream application (**Section 3.4.1**), we designated seven variables associated with 62 unique levels ($S_0 = 7, S = 62$), primarily socio-demographic and activity-related attributes, as SB. The remaining five trip-context related variables, encompassing 21 unique categorical levels ($E = 5, Z = 21$), were assigned as EB, representing factors that may vary across trips or contexts for the same individual. Brief descriptions of all 15 explanatory variables and the surveyed choice profile are summarized in **Table 4**, which also provides variable types (survey coding) and key level definitions used for subsequent modeling experiments.

Table 4 Explanatory variables in the filtered SwissMetro dataset and their assigned roles under the practical variable-role strategy (N=9,036)

Variable	Description	Type	Assigned role for the manual split
TT	Door-to-door mode-specific travel times in minutes, scaled by 1/100	Continuous	LIP
CO	Mode-specific travel costs in CHF, scaled by 1/100	Continuous	LIP
HEADWAY	Headways for non-car modes in minutes, scaled by 1/100	Continuous	LIP
AGE	Traveler's age class (5 levels)	Categorical (Ordinal)	SB
MALE	Traveler's gender (2 levels)	Categorical (Binary)	SB
INCOME	Traveler's income class (5 levels)	Categorical (Ordinal)	SB
GA	Whether the traveler holds the annual seasonal ticket (2 levels)	Categorical (Binary)	SB
PURPOSE	Trip purpose (9 levels, including business, leisure, shop, work)	Categorical	SB
ORIGIN	The trip's originating region or Canton (18 levels)	Categorical	SB
DEST	The trip's destined region or Canton (21 levels)	Categorical	SB
FIRST	Recorded train ticket class (first/not) of the survey respondent (2 levels)	Categorical (Binary)	EB
TICKET	The ticket type for the trip (9 levels: e.g., one-way, discounted)	Categorical	EB
PAIDBY	Who paid for the trip (4 levels: self, employer, mixed, unknown)	Categorical	EB
LUGGAGE	Luggage counts of the traveler for the trip (3 levels: none, one, several)	Categorical (Ordinal)	EB
SEATS	Seat configuration for hypothetical SM mode (3 levels with a non-SM dummy)	Categorical	EB
CHOICE	TRAIN (8.6%), CAR (34.1%), SM (57.3%)	-	Choice (y)

5.2. DCM-SEAL Results on SwissMetro Data with the Fixed Variable Roles

We first estimated models using three configurations of variable-role assignment, in which (1) all twelve non-core categorical variables were allocated entirely to EB, (2) all twelve entirely to SB, or (3) split using the manual role configuration described in **Table 4**. As in **Section 4**, these three settings correspond respectively to the E-MNL, the neural-membership latent class choice model (NN-LCCM), and the proposed DCM-SEAL specification. The DCM-SEAL specification's practical split, as discussed earlier, reflects a strategy that non-expert modelers can readily adopt to develop interpretable and reusable models: variables closely tied to individuals (e.g., sociodemographic or activity-related features such as origin, destination, and purpose) are assigned to the segmentation network to define behaviorally heterogeneous latent classes, while trip-contextual variables are embedded to capture within-class marginal taste.

Following the procedure in **Section 4.3**, we conducted 200 Optuna trials for each configuration, using three latent-class scenarios ($C = 2, 3, 4$) for both NN-LCCM and DCM-SEAL, resulting in a total of seven model specifications. For each, we selected the hyperparameter combination that achieved the highest test-set ρ^2 (see **Appendix C** for details). Using these best-performing settings, each model was then retrained repeatedly until 30 runs exhibited test ρ^2 values exceeding the 95th percentile of their respective Optuna-generated 200-trial distributions. **Table 5** summarizes the resulting model fit statistics across the seven configurations, highlighting the comparative predictive performance of each framework.

Table 5 SwissMetro model performance comparison across variable-role configurations and the number of latent class specifications

Models (selected 30 runs)	Number of training trials required to obtain 30 good runs	Test ρ^2 across the 30 runs			Test log-likelihood across the 30 runs		
		Mean	Std. Deviation	Max	Mean	Std. Deviation	Max
E-MNL	445	0.3540	0.0004	0.3550	-1283.2	0.89	-1281.3
NN-LCCM ($C = 2$)	263	0.3757	0.0066	0.3945	-1239.5	13.05	-1202.1
NN-LCCM ($C = 3$)	240	0.3972	0.0116	0.4312	-1197.4	23.01	-1129.8
NN-LCCM ($C = 4$)	336	0.4204	0.0080	0.4431	-1151.2	15.98	-1106.2
DCM-SEAL ($C = 2$)	328	0.3978	0.0069	0.4173	-1191.6	13.67	-1156.8
DCM-SEAL ($C = 3$)	179	0.4264	0.0063	0.4456	-1139.3	12.57	-1101.3
DCM-SEAL ($C = 4$)	243	0.4482	0.0049	0.4628	-1096.0	9.76	-1067.0

The results show a consistent improvement in both test log-likelihood and ρ^2as the model transforms from E-MNL to NN-LCCM and finally to DCM-SEAL. Comparing the latent class models (NN-LCCM and DCM-SEAL), incorporating DCM-SEAL's embedding component yields notable gains, particularly for higher latent-class dimension ($C = 3$,4), where the mean test ρ^2 improves by approximately 0.02-0.03 relative to NN-LCCM. Also, the relatively smaller standard deviations of DCM-SEAL indicate stable convergence and robustness of the optimized hyperparameter settings given our model selection criteria. Overall, DCM-SEAL achieves the best predictive fit with $C = 4$ ($\rho^2 = 0.448$), demonstrating its enhanced ability to model intricate travel behavior. On the other hand, both NN-LCCM and DCM-SEAL exhibit improved accuracy as the number of latent classes increases. However, consistent with the previous studies (Kim and Mokhtarian, 2023; Mouter et al., 2017), arbitrarily increasing C is not recommended. Although predictive accuracy continues to improve with additional classes, excessive segmentation complicates behavioral interpretation, increases the risk of overfitting, and diminishes the model's practicality for policy analysis.

Complementary to **Table 5**, **Table 6** provides detailed comparisons of parameter estimates, standard errors, and model fit metrics juxtaposed with previously reported results from Sifringer et al. (2020)'s L-MNL and benchmark MNL, and Arkoudi et al. (2023)'s E-MNL[4]. Other studies that used SwissMetro and a data-driven DCM are excluded if they allowed interactions between LIP and non-core variables in the utility formulation that do not guarantee a similar level of interpretation for comparability, such as ResNet (Han et al., 2022), the interacting variant of L-MNL, and Arkoudi et al. (2023)'s EL-MNL. In addition, since no comparable prior results exist for 2-class NN-LCCM on this dataset, we included its performance using the $E = 0$ configuration trained on our DCM-SEAL framework for reference. Finally, the DCM-SEAL model with the highest test-ρ^2 among the 30 runs was selected as *the focal model*. Its model evaluation metrics are also presented in **Table 6** and serve as the basis for the subsequent discussions on model interpretation and practical applications.

It is worth noting that several points were considered in reporting and benchmarking results in **Table 6**. First, unlike the synthetic data experiment, the SwissMetro dataset lacks known ground-truth parameters and class structures; thus, we report the average values of estimated coefficients (β's), not MRSs, with their standard errors, computed as the sample standard deviation of the 30 runs divided by $\sqrt{30}$, along with associated t-statistics and p-values under the null hypothesis $\beta = 0$. Among all models, the benchmark MNL results from Sifringer et al. (2020) serve as a reliable analytical reference given their stability and established estimation procedure.

Second, because parameter magnitudes may vary across models with differing scale normalizations, robust t-statistics and p-values are preferred for cross-model interpretation rather than raw β estimates. In addition, for each model, we report the Value of Time (VoT), computed as the ratio of the estimated travel-time coefficient ($\hat{\beta}_{TT}$) to the cost coefficient ($\hat{\beta}_{CO}$), disaggregated by latent class where applicable. Finally, due to differences in model specification and reporting focus across previous studies (e.g., some used one-hot while others used embeddings), the model comparison is restricted to the three core-LIP coefficients and selected others that are semantically consistent with our DCM-SEAL outputs.

Third, for models with multiple latent classes ($C > 1$), the ordering of latent classes is inherently arbitrary without ground-truth membership labels as an external reference. Therefore, classes were reindexed post hoc such that higher-numbered classes correspond to higher VoT estimates. Yet, this sorting does not imply behavioral equivalence across 30 model runs for the same-numbered classes, since membership probability for the same individual across runs can differ. Consequently, to ensure interpretability and comparability, only the binary latent-class configuration ($C = 2$) for NN-LCCM and DCM-SEAL is reported in **Table 6**, while results for higher-class models are discussed later in **Section 5.3**.

The DCM-SEAL parameter estimates indicate that both classes' results exhibit the expected negative signs for core variables, reaffirming the model's adherence to behavioral realism. In addition, even though latent class identification can vary substantially during optimization, most coefficients remain statistically significant—only the alternative-specific constant (ASC) of Class 1 for SM is not significantly different from zero based on the p-values. Meanwhile, these small parameter variations observed across different initialization seeds do not imply that the model is unreliable. In neural-embedded latent class architectures, different random seeds will lead the optimizer to converge to different, but almost equally valid (given comparable log-likelihood values), local optima. Behaviorally, this suggests that there is no single, unique way to partition a complex "latent" travel segments. Instead, different initializations reveal alternative, mathematically sound dimensions of behavioral heterogeneity, highlighting different ways to group travelers and understand their average modal preferences based on the input data.

The resulting VoT estimates are stable across multiple runs, and the aggregated estimates fall within the empirically plausible range reported in prior SwissMetro studies (1.13 CHF/min for L-MNL to 1.92 CHF/min for MNL). Using the dominant Class 1 (covering, on average, 76% of samples with the estimated Class 1 membership probability or $\hat{\pi}^1 > 0.5$), the VoT computed as $\hat{\beta}^1_{TT}/\hat{\beta}^1_{CO}$ equals 1.53 CHF/min, while Class 2 exhibits a substantially higher VoT of 6.07 CHF/min. This segmentation reveals two traveler groups: one with relatively moderate sensitivity to travel time and cost, and another that places a disproportionately high value on time savings. Such elevated VoT values, particularly for the high-VoT segment, are consistent with Han et al. (2022) , who found that gradually incorporating more interaction structures in the SwissMetro dataset consistently increases the estimated average VoT, suggesting that more flexible utility representations capture stronger time-valuation behaviors.

On the other hand, the ASC patterns reveal distinct preference structures across classes. The lower-VoT segment (Class 1) prefers car travel over train. In contrast, the difference between train and the new SM mode is statistically insignificant, as are the differences between train and car (which are likely to share many survey-unobserved modal properties, captured by ASCs). In contrast, those in the higher-VoT segment (Class 2) show a clear preference hierarchy of SM > Train > Car. These results illustrate how class-specific parameters can guide policymakers and service providers in formulating differentiated strategies; for example, premium SM offerings tailored to time-sensitive travelers versus cost-sensitive incentives for car users. This observation also aligns with Han et al. (2022), who estimated separate travel-time coefficients for the three travel modes and found, consistently across all four model variants, that the car mode exhibited the lowest VoT.

Compared to NN-LCCM, DCM-SEAL achieves higher prediction accuracy, similar behavioral directionality, and notably greater flexibility in interpreting non-core variables, which can also be used in future applications such as agent-based simulation studies. This supports the premise that DCM-SEAL's integrated segmentation and embedding mechanisms enhance both accuracy[5] and practicability. Overall, while other benchmark models serve as valuable reference points, DCM-SEAL demonstrates a balanced trade-off between interpretability and predictive accuracy, delivering class-specific behavioral differentiation without compromising econometric interpretability. Across all comparisons, model accuracy metrics (test log-likelihood, ρ^2, and CEL) show that even the binary-class DCM-SEAL performs on par with or better than comparable data-driven DCMs. Moreover, while L-MNL's non-core variables contribute to utility through less interpretable DNN transformations (as a standalone complementary to the error term), conventional MNL underperforms predictive measures, and DCM-SEAL consistently offers interpretable and accurate insights.

[4] We followed, as closely as possible, the data preprocessing and reporting protocols (i.e., 30-run averages) adopted by the L-MNL and E-MNL studies. However, because each study applied its own random 80/20 training-test split, the resulting subsets are not identical. Consequently, performance metrics such as log-likelihood and ρ^2 are not directly comparable across studies; they rather provide a rough benchmark for gauging typical model behavior. For instance, if a particular test set contains a higher proportion of hard-to-predict observations, its reported performance will naturally appear worse than that of another study with easier test cases. This issue is further compounded by the stochastic nature of DNN-based training, meaning that independent evaluators may obtain higher-accuracy models through more extensive hyperparameter tuning or by random variation in optimization outcomes for any non-MNL models reported in **Table 6**.

[5] The direct accuracy comparison between Arkoudi et al. (2023)'s E-MNL and our DCM-SEAL ($C = 2$) is inconclusive based solely on the reported test log-likelihood values in **Table 6**. We attempted to reproduce a comparable E-MNL result using our framework; however, several methodological differences prevented a direct replication. These include variations in dataset splits (as previously noted), optimization algorithms (Adam versus AdamW), representations of core-LIP variables (convolutional neural network versus direct linear terms), regularization and normalization strategies, and reporting conventions. Nevertheless, two observations are noteworthy: (1) DCM-SEAL with more than two latent classes ($C > 2$) achieved substantially higher test log-likelihood than Arkoudi et al. (2023)'s E-MNL, and (2) the E-MNL re-estimated under our framework (line 1 of **Table 5**) yielded lower log-likelihood than the corresponding DCM-SEAL models. Together, these results suggest that, in general, DCM-SEAL is likely to attain higher predictive accuracy than E-MNL with the same dataset.

Table 6 Parameter estimates and behavioral indicators of DCM-SEAL and benchmark models on the SwissMetro dataset

Model-wide Statistics	Comparable Parameters	Estimates	Std. errors	*t*-statistics	*p*-value
MNL (Sifringer et al., 2020: analytic benchmark results)	ASC_Car	1.08	0.162	6.67	0.00
Test[a] log-likelihood: -1,433	ASC_SM	1.05	0.153	6.84	0.00
Test ρ^2: 0.28	CO	-0.695	0.042	-16.4	0.00
Test CEL: Not provided	HE	-0.733	0.113	-6.47	0.00
VoT: 1.92 CHF/min	TT	-1.34	0.051	-26.2	0.00
L-MNL (Sifringer et al., 2020; 30 runs average)	ASC_Car	0.106	0.174	0.61	0.54
Test log-likelihood: -1,181	ASC_SM	0.454	0.163	2.80	0.01
Test ρ^2: 0.41	CO	-1.38	0.048	-28.5	0.00
Test CEL: Not provided	HE	-0.860	0.127	-6.77	0.00
VoT: 1.13 CHF/min	TT	-1.56	0.056	-28.0	0.00
E-MNL (Arkoudi et al., 2023; 30 runs average)	ASC_Car	-0.172	0.125	-1.38	0.167
Test log-likelihood: -1,197	ASC_SM	-0.461	0.107	-4.31	0.000
Test ρ^2: Not provided	CO	-1.11	0.047	-23.6	0.000
Test CEL: Not provided	HE	-0.80	0.129	-6.19	0.000
VoT: 1.27 CHF/min	TT	-1.41	0.054	-25.9	0.000
	FIRST [b]	0.91	0.223	4.08	0.000
	TICKET [b]	1.75	0.071	24.7	0.000
	PAIDBY [b]	0.941	0.108	8.73	0.000
	LUGGAGE [b]	0.668	0.120	5.59	0.000
	SEAT [b]	0.624	0.203	3.08	0.002
NN-LCCM ($C = 2$; 30 runs average)	Class 1: ASC_Car	1.531	0.097	15.9	0.000
Test log-likelihood: -1,240	Class 1: ASC_SM	0.456	0.127	3.59	0.001
Test ρ^2: 0.38	Class 1: CO	-1.108	0.095	-11.7	0.000
Test CEL: 0.6859	Class 1: HE	-1.416	0.089	-16.0	0.000
VoT (Class 1): 1.70 CHF/min	Class 1: TT	-1.884	0.055	-34.2	0.000
Average proportion of samples inclined to Class 1: 73%	Class 2: ASC_Car	-2.279	0.135	-16.9	0.000
VoT (Class 2): 10.80 CHF/min	Class 2: ASC_SM	1.280	0.093	13.8	0.000
Average proportion of samples inclined to Class 2: 27%	Class 2: CO	-0.059	0.003	-20.0	0.000
	Class 2: HE	-0.325	0.039	-8.33	0.000
	Class 2: TT	-0.637	0.066	-9.62	0.000
DCM-SEAL ($C = 2$; 30 runs average)	Class 1: ASC_Car	0.579	0.153	3.79	0.001
Test log-likelihood: -1,196	Class 1: ASC_SM	-0.158	0.171	-0.92	0.364
Test ρ^2: 0.40	Class 1: CO	-1.244	0.110	-11.3	0.000
Test CEL: 0.6619	Class 1: HE	-0.799	0.072	-11.1	0.000
VoT (Class 1): 1.53 CHF/min	Class 1: TT	-1.910	0.074	-25.8	0.000
Average proportion of samples assigned to Class 1: 76%	Class 1: FIRST	0.632	0.081	7.83	0.000
VoT (Class 2): 6.07 CHF/min	Class 1: TICKET	1.038	0.132	7.87	0.000
Average proportion of samples assigned to Class 2: 24%	Class 1: PAIDBY	0.894	0.083	10.7	0.000
	Class 1: LUGGAGE	0.841	0.077	10.9	0.000
Focal model's test log-likelihood: -1,157	Class 1: SEAT	0.923	0.088	10.5	0.000
Focal model's test ρ^2: 0.42	Class 2: ASC_Car	-0.686	0.174	-3.95	0.001
Focal model's test CEL: 0.6401	Class 2: ASC_SM	0.765	0.173	4.42	0.000
Focal model's VoT (Class 1): 0.99 CHF/min	Class 2: CO	-0.152	0.037	-4.11	0.000
Focal proportion of samples inclined to Class 1: 80%	Class 2: HE	-0.681	0.066	-10.4	0.000
Focal model's VoT (Class 2): 6.33 CHF/min	Class 2: TT	-0.925	0.106	-8.74	0.000
Focal proportion of samples inclined to Class 2: 20%	Class 2: FIRST	0.985	0.106	9.30	0.000
	Class 2: TICKET	1.497	0.118	12.7	0.000
	Class 2: PAIDBY	0.939	0.089	10.6	0.000
	Class 2: LUGGAGE	1.046	0.110	9.52	0.000
	Class 2: SEAT	1.210	0.128	9.49	0.000

[a] See **Footnotes 3** and **4** for essential caveats regarding the comparability of accuracy metrics across models.

[b] The five variables (FIRST, TICKET, PAIDBY [WHO], LUGGAGE, SEAT) from E-MNL are included to compare the stability of DCM-SEAL's corresponding estimates via their t-statistics. However, it should be noted that, because an EB variable's total effects are computed as compound terms (**Figure 1**), direct comparison of coefficient magnitudes between the two models is not meaningful.

Beyond the predictive accuracy and econometric insights for the core variables, the following subsections illustrate how DCM-SEAL bridges the two extremes of its foundational models—E-MNL and NN-LCCM—by combining their respective strengths. The discussion is based on the focal model, which achieved the highest test ρ^2 (0.417) among the 30 DCM-SEAL runs given $C = 2$. The analysis examines whether introducing additional complexity (a component) from either end is justified to achieve the integrated DCM-SEAL architecture. In essence, while E-MNL lacks segmented behavioral interpretation linked to non-core interactions, and NN-LCCM offers no direct interpretability of individual categorical levels' effects, DCM-SEAL provides both class-specific segmentation and interpretable utility contributions, striking a practical balance between analytical transparency and modeling flexibility.

5.2.1. Discussion of the focal DCM-SEAL model's segmentation

For clearer interpretation, the two latent classes identified from the focal model were further divided into three subgroups based on each choice scenario's model-predicted probability of belonging to Class 2 or $\hat{\pi}^2$, which is $1 - \hat{\pi}^1$. Specifically, these groups represent: (1) deterministically *low VoT*, (2) *indecisive*, and (3) deterministically *high VoT*. The cutoff thresholds were set at 15% and 85% based on the computed Class 2 membership probability or $\hat{\pi}^2$, resulting in 2,448 (27.1%), 5,409 (59.9%), and 1,179 (13.0%) observations assigned to each subgroup, respectively. **Figure 4** illustrates the mode choice probabilities across the three subgroups. The results show that the low-VoT subgroup is more inclined to choose the car mode, whereas the high-VoT subgroup predominantly selects non-car alternatives. This pattern is consistent with the class-specific ASC estimates presented in **Table 6**, and aligns with previous studies' findings by Han et al. (2022) and Kim and Bansal (2024).

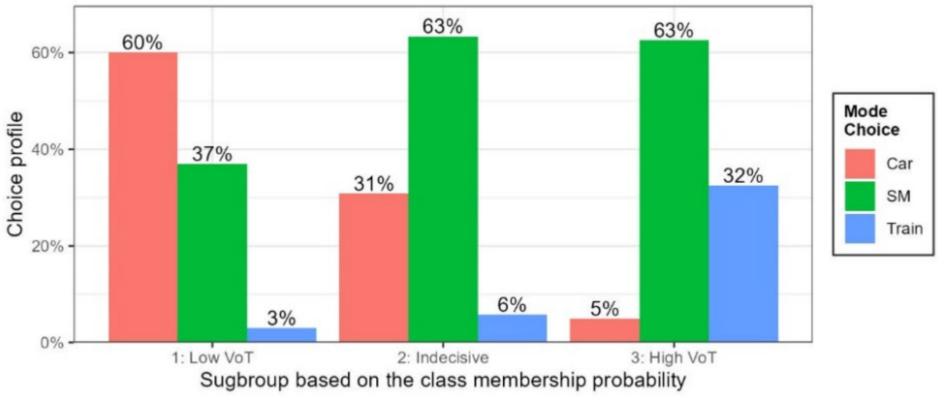


Figure 4 The focal model's mode choice profile by subgroup

Subsequently, we examined the subgroup composition by SB variables, excluding ORIGIN and DESTINATION. **Figure 5** summarizes the distributions for PURPOSE, AGE, MALE, INCOME, and GA. For example, among all 7,182 observations where MALE is "yes", 1,944 (27%), 4,419 (62%), and 819 (11%) are classified into Subgroups 1-3, respectively; the numeric labels denote these proportions. Note that all four "returning (to home) from X" minor categories were consolidated into a single level (297 total) for simplicity.

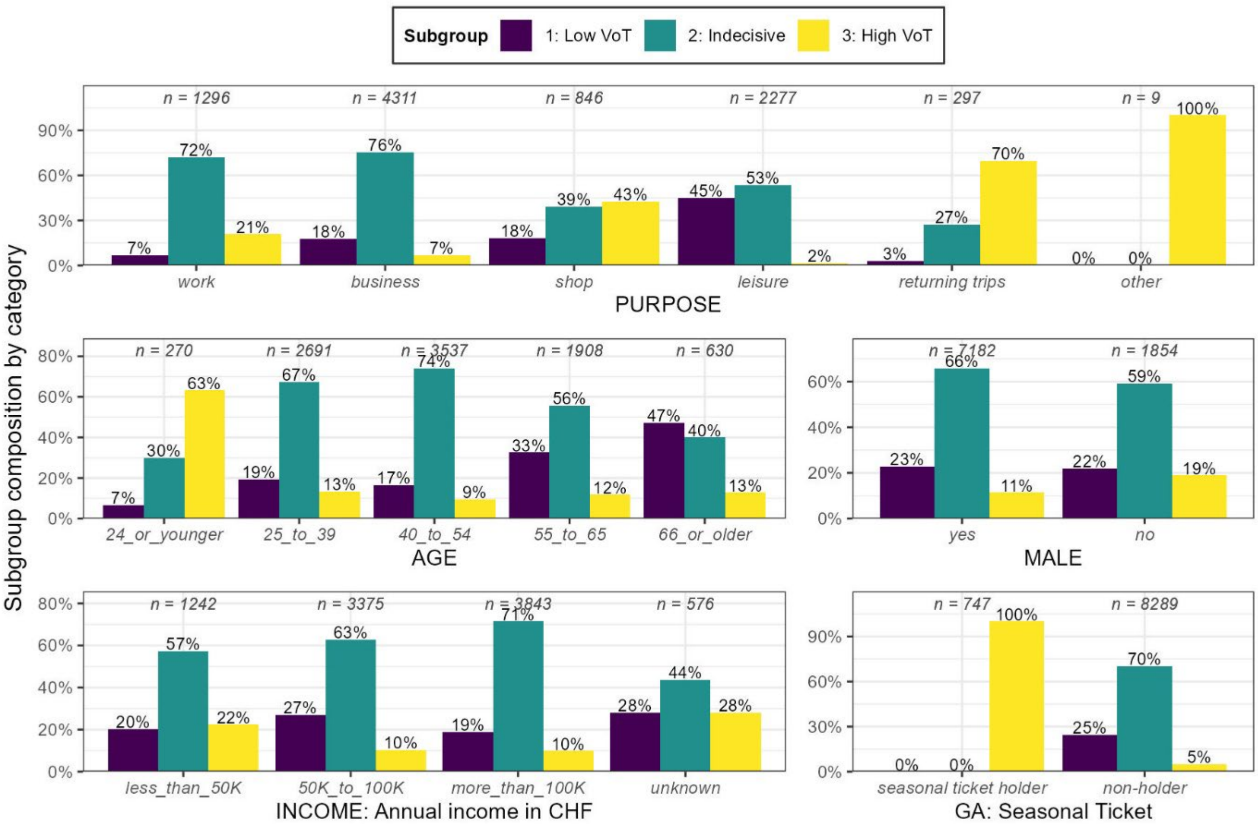


Figure 5 The focal model's subgroup composition by each categorical level of the five segmentation bases (each x-axis tick's sum=1)

The result reveals a non-conventional segmentation pattern. Typically, we can expect that middle-aged individuals with higher incomes who travel primarily for work or business are more likely to be classified as the high-VoT subgroup. Interestingly, the segmentation shows that shopping trips—apart from the two minor categories "returning trips" and "other" (each <5%)—are associated with higher VoT than work or business trips. This finding likely reflects the nature of the SwissMetro dataset, which focuses on intercity travel: "*inter-regional* shopping" trips typically involve high-importance activities and thus reflect stronger time sensitivity, or occur only when there is no financial burden.

More likely, this purpose-related non-conventionality could be related to the GA or the Swiss nationwide seasonal ticket. The focal model's membership network deterministically classifies GA holders (with 100% probability) into the high-VoT subgroup, while non-holders are primarily assigned to the low- or indecisive VoT groups. This result aligns with previous studies' findings (Han et al., 2022; Kim and Bansal, 2024) that GA ownership significantly shapes travel behavior: since GA holders have already paid a high, fixed cost for the seasonal ticket purchase, their subsequent mode choices depend primarily on travel time rather than monetary cost, leading naturally to higher VoT estimates.

The strong influence of GA also complicates the interpretation of AGE and INCOME, as these variables interact nonlinearly within the membership network. **Table 7** presents normalized contingency tables between (1) GA and AGE and (2) GA and INCOME, where each column is scaled to sum to one or 100%. The results show that younger respondents *24_or_younger* (30%) and lower-income or income-unknown groups (12% and 14%, respectively) have disproportionately high GA ownership compared to the overall sample average (8%). Consequently, these groups, typically not associated with high VoT in conventional interpretations, are predicted to be more likely to belong to the high-VoT subgroup due to their strong interaction with GA ownership. This highlights DCM-SEAL's ability to uncover cross-variable interactions that are difficult to capture with traditional linear segmentation.

Table 7 Normalized contingency table for GA against AGE and INCOME from the filtered SwissMetro dataset (N=9,036)

GA	**Overall Average**	**INCOME (1,000 CHF/year)**				**AGE**				
		<50	**50- 100**	**>100**	**Unknown**	**<25**	**25-39**	**40-54**	**55-65**	**>65**
Holds GA	8.3%	**11.6%**	6.9%	7.5%	**14.1%**	**30.0%**	8.4%	7.6%	6.6%	7.1%
Does not hold GA	91.7%	88.4%	93.1%	92.5%	85.9%	70.0%	91.6%	92.4%	93.4%	92.9%

Nevertheless, it is essential to reiterate that marginal interpretations like **Figure 5** by individual SB variables/levels must be made cautiously. Except for highly deterministic categorical levels, such as *holds* GA, or sparse levels like PURPOSE: *other*, class membership probabilities result from deep, multivariate interactions among all SB inputs, including ORIGIN and DESTINATION (omitted in the figure for clarity). Nevertheless, this decomposition provides valuable insights into how latent classes are composed and into the characteristics that distinguish individuals who are deterministically assigned from those near the decision boundary in the latent segmentation space.

5.2.2. Discussion of the focal DCM-SEAL model's embedding

One way to interpret the embedding component in terms of utility is by computing the total contribution of each categorical level in the EB. This contribution is obtained by multiplying the unit embedding vector (each row of the class-specific embedding weight matrix, $\mathbf{M}^c$) by the corresponding positive parameter estimate, β_e^{ci}. The resulting value represents both the direction and magnitude of each categorical level's influence on the utility, differentiated by class and by alternative. **Figure 6** illustrates the class- and alternative-specific utility contributions computed from the focal model for the two EB variables (PAIDBY, LUGGAGE) that are generic across all three alternatives. As noted earlier, these EB effects are also conditioned by SB interactions through the latent-class structure. For instance, some levels of TICKET (in EB)—omitted from the figure because they do not apply to the Car mode—are strongly correlated with GA (in SB), as GA holders do not pay full fares.

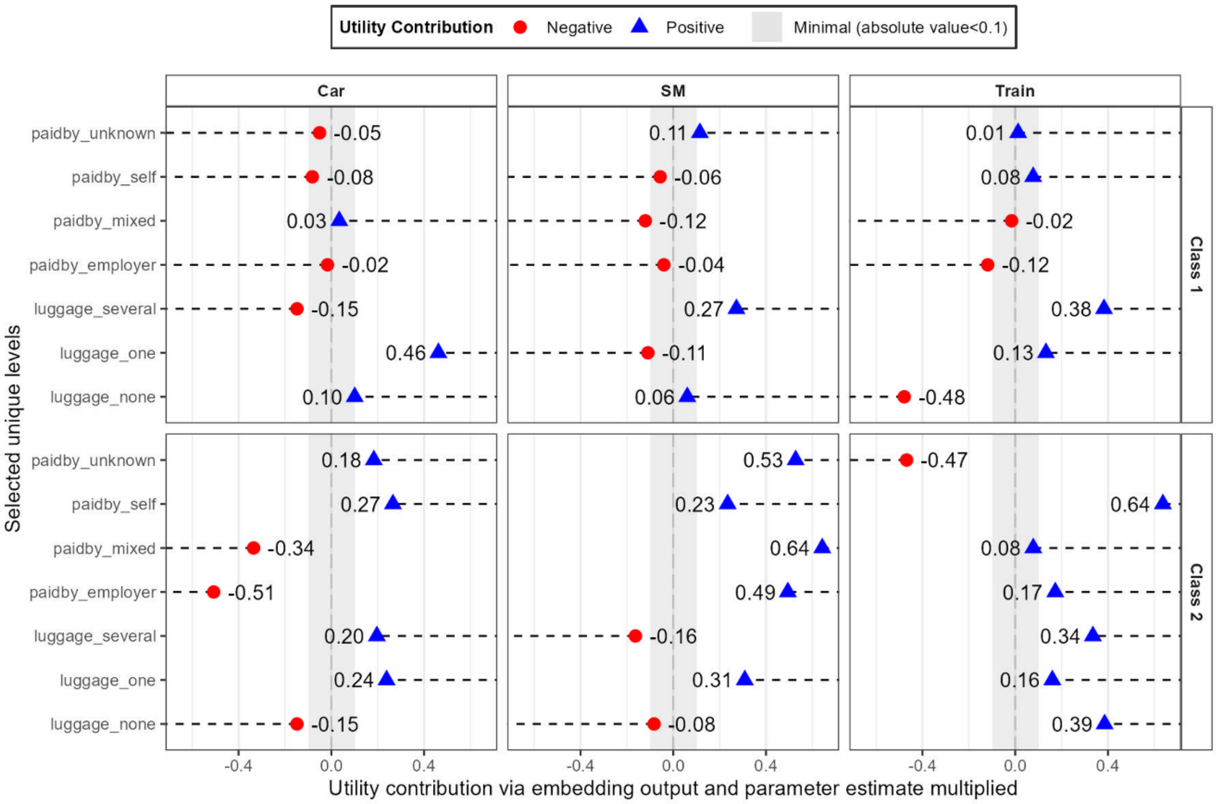
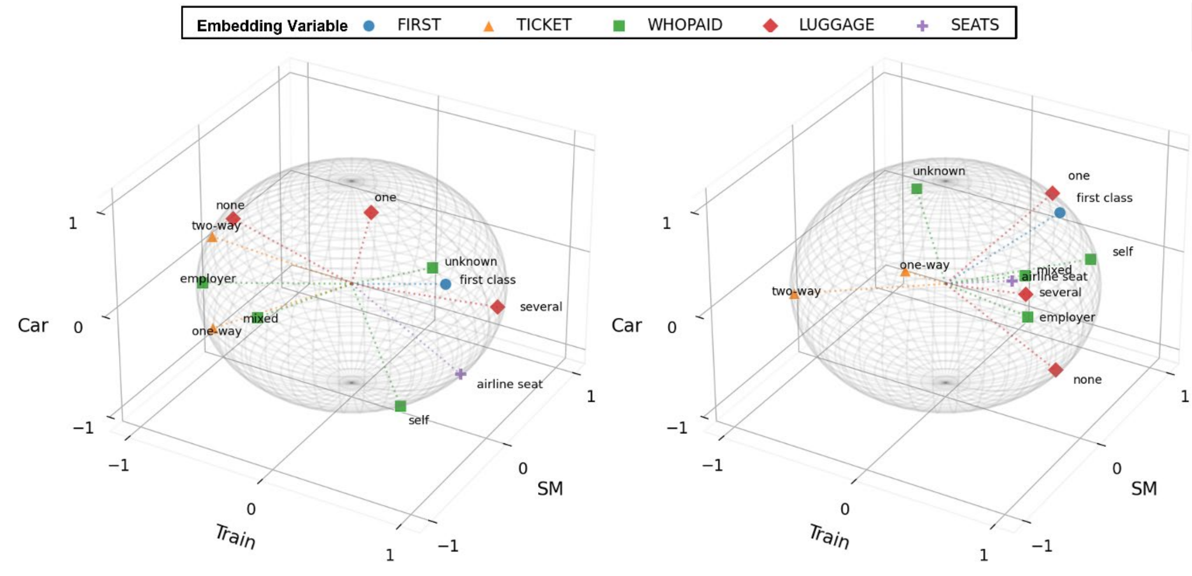


Figure 6 Class- and alternative-specific utility contributions of the focal model's selected variables in embedding base

While these categorical levels' effects are conceptually similar to the conventional one-hot-encoded levels' associated parameters in an MNL framework, their class-specific differentiation reveals LCCM and DCM-SEAL's ability to capture heterogeneous behavioral patterns that would otherwise be averaged out in aggregate models. In particular, the direction and magnitude of each EB level's contribution vary between Class 1 and Class 2, reflecting distinct preference structures even for the same attribute categories. For example, while self-funded and employer-reimbursed trips have minimal influence on utility for Class 1, their effects diverge sharply in Class 2, where the contributions of these expense sources differ substantially between train and non-train modes. Overall, it is observed that the utility impact from each category level is generally higher (in absolute value) for Class 2, and for non-car modes. We redirect interested readers to Arkoudi et al. (2023) for a detailed behavioral interpretation of these utility contributions with embedding in the aggregate (non-segmented) context. Instead, the following discussion highlights the distinctive advantage of the embedding component—its learnable weight vectors constrained on the unit hypersphere—which go beyond simple one-hot encoding representations to capture cross-level relational structures. In addition, this is enhanced by the class-specific representation learning inherent to the DCM-SEAL framework.

Figure 7 visualizes the class-specific embedding weight vectors for selected, behaviorally meaningful levels (i.e., excluding some ticket types affected by GA and keeping only one level for binary variables), mapped onto the unit hypersphere in a latent space where each axis corresponds to the modal utility dimension (Train, SM, and Car). The plots reveal that both the absolute positions of the embedding vectors and their pairwise angular relationships differ substantially across the two latent classes, indicating distinct behavioral representations learned for each segment.



(a) Class 1 (low VoT) (b) Class 2 (high VoT)

Figure 7 The focal model's class-specific embedding weight vectors for each category level, visualized on the unit sphere

As a complementary analysis, we computed pairwise angular distance or cosine similarities (ranging from 0 for orthogonal vectors to ±1 for perfectly collinear ones) among the 11 unit embedding vectors from the focal model shown in **Figure 7**, producing a total of 55 pairwise relationships per class (the full computation result is provided in **Appendix D**). Because these vectors are normalized to unit length, their directional similarity can be directly assessed using the dot product of their coordinate vectors (a row vector), which equals the cosine of the angle between them. According to Arkoudi et al. (2023), categories whose vectors are nearly collinear (absolute cosine near 1) indicate overlapping behavioral information and may thus be merged or omitted to achieve a more parsimonious utility specification. Conversely, orthogonal vectors (cosine near 0) capture complementary behavioral effects, providing richer explanatory power.

Table 8 summarizes these relationships by reporting, for each categorical level, the mean absolute cosine similarity within each class (indicating within-class information overlap) and the between-class cosine similarity (indicating how consistently each level behaves across latent classes). This explains how each categorical level contributes to the behavioral information within and between the two identified segments.

Table 8 Average absolute intra-class and pairwise raw between-class cosine similarities of embedding weight vectors from the focal model

Variable	Level	Mean absolute cosine (Class 1)	Mean absolute cosine (Class 2)	Between-class same-level comparison	
				Cosine	Degree
FIRST	*first class*	0.500	0.498	0.895	26°
TICKET	*one-way*	0.490	0.547	0.493	60°
TICKET	*two-way*	0.629	0.544	0.264	75°
LUGGAGE	*none*	0.566	**0.379**	-0.979	168°
LUGGAGE	*one*	**0.285**	0.422	0.155	81°
LUGGAGE	*several*	0.645	0.541	0.470	62°
SEATS	*airline seat*	0.539	0.510	0.169	80°
PAIDBY	*employer*	0.626	0.462	-0.130	97°
PAIDBY	*mixed*	0.426	0.535	-0.349	110°
PAIDBY	*self*	0.412	0.507	-0.985	170°
PAIDBY	*unknown*	0.430	0.468	0.099	84°

The results suggest that, within Class 1, the LUGGAGE: *one* level provides the most distinctive contribution to the utility computation, whereas LUGGAGE: *none* does so for Class 2. Conversely, LUGGAGE: *several* in Class 1 and TICKET: *one-way* in Class 2 exhibit the highest average internal alignment with other levels, indicating that their behavioral influence may be partially shared. Notably, the range of mean absolute cosine values is wider for Class 1 (0.285-0.645) than for Class 2 (0.379-0.547), implying that the embedding vectors operate more diversely (orthogonally) within Class 1's latent utility space—may indicate Class 2's behavior is more governed by out-of-EB variables derived from the SB interaction and/or the core LIP given their high VoT. On the other hand, the pairwise between-class cosine similarities provide insight into how consistently the same categorical levels behave across latent segments. For example, PAIDBY: *self* exhibits almost opposite (170°) directional behavior between the two classes, whereas the FIRST: *first class* indicator shows the highest consistency (26°) across them.

Beyond the observations of Arkoudi et al. (2023), we note that high (absolute) cosine similarity between embedding vectors does not always imply information overlap. When collinearity occurs *within* a single categorical variable, consolidating the levels can enhance both model parsimony and interpretability. For example, the nearly linear PAIDBY: *employer* and PAIDBY: *mixed* within-variable pair in Class 2 (cosine = 0.943) could feasibly be merged without substantial behavioral loss. However, when strong alignment across variables arises, it may instead reflect behavioral consistency rather than duplication. Because each embedding variable contributes additively to total utility (**Equation (8)**), between-variable alignment (such as between PAIDBY: *mixed* and SEAT: *airline seat* for Class 2, with a cosine value of 0.994) can indicate complementary behavioral tendencies that reinforce rather than substitute for one another.

In addition, it remains uncertain how removing a level identified as collinear would alter the geometry of the remaining embeddings, both those that initially exhibited similar directional orientations and others that may be implicitly correlated through higher-order interactions (e.g., with SB interactions). Such potential structural dependencies imply that pruning levels could reshape the learned embedding space in non-trivial ways. Nevertheless, maintaining embedding vector pairs that are mutually orthogonal as many as possible (e.g., LUGGAGE: *none* and SEAT: *airline seat* in Class 2 with the cosine value 0.040) yields a behaviorally rich representation.

Therefore, the mean cosine values reported in **Table 8** may serve as a practical diagnostic to refine the model, ensuring that each categorical level contributes distinct, non-redundant information. In summary, together with the membership neural network (which effectively captures inter-variable interactions), the DCM-SEAL framework demonstrates that correlated or interacting effects are primarily handled through the segmentation mechanism. At the same time, the embedding component adds additional behavioral nuance to each segment. This synergy between the EB and SB structures underscores the methodological advancement and interpretive depth of the DCM-SEAL architecture.

5.3. Grid Search Over Configurations: EB-SB Variable Role Split and The Number of Latent Classes

As discussed in the previous section, splitting non-core variables into SB and EB has meaningful implications for both interpretability and performance. Also, given a particular dataset, we can assume certain categorical variables or specific combinations of them may function more effectively within EB, while others may serve better as SB inputs. In the synthetic data experiment (**Section 4**), the ground-truth variable split was known by design. In contrast, in the SwissMetro experiment in this section, we used a manual split intended for a downstream application of an activity-based modeling simulation. Nonetheless, even under a consistent modeling objective, some variables fall into a conceptual gray area. For instance, origin, destination, and trip purpose were categorized as SB variables in our experiment under the assumption that individual activities and locations are predetermined. Alternatively, one could assume that activities themselves are stochastic elements within a simulation and not fixed to individual agents, in which case it is more practical to treat these as EB. In addition, such ambiguity naturally varies when the modeling purpose or downstream application changes, meaning that the EB-SB division ultimately depends on the analyst's judgment, until an algorithm is developed in the future to systematically find a combination that best suits the incumbent objective—for example, using intermediate metrics derived from results like **Figure 5** and **Table 8** for a specific run and inform the algorithm to proceed towards an effective direction. This section, therefore, examines how different variable splits influence model accuracy, exploring whether an "optimal" configuration exists solely from the standpoint of predictive performance for a given dataset.

This exhaustive procedure, referred to as the *grid search*, tested a total of 12,286 *configurations*. The initial set of configurations was derived from all possible *splits* of the 12 categorical variables in the SwissMetro dataset (**Table 4**) into two mutually exclusive and exhaustive subsets representing the EB and SB (i.e., the variable role assignment). This yields $2^{12} = 4{,}096$ possible splits assigned ID from 1 to 4096 (1: all-SB, 4096: all-EB). Each split was then evaluated under three latent-class specifications ($C = 2,3,4$). However, the extreme case in which all 12 variables are assigned to EB and none to SB (ID: 4096) simplifies the model into E-MNL with fixed $C = 1$ without the C variations; therefore, the total number of valid configurations becomes $4{,}095 \times 3 + 1 = 12{,}286$. These EB/SB splits and the number of latent classes C are neither model parameters (e.g., coefficients, weights, or biases) nor hyperparameters optimized by Optuna (e.g., learning rate γ, regularizations). Rather, they represent configuration parameters, which are exogenous choices that must be explored manually to identify effective model specifications.

For each configuration, using the same hyperparameter search space defined in **Table 1**, we conducted 50 Optuna trials to tune hyperparameters, then retained only the top 10% "reasonably tuned" trials based on the test set ρ^2. This process resulted in 61,430 model runs ($12{,}286 \times 50 \times 0.1$), each corresponding to a distinct configuration (a combination of the split and the number of latent classes) and at least moderately tuned hyperparameter settings given the configuration, which are collectively compared and analyzed in this section.

First, to investigate whether certain variable-role splits consistently yield superior performance, we examined the occurrence frequency of each split among the top 300 model runs (test-set $\rho^2 > 0.446$) out of all 61,425 runs, excluding the five all-EB cases for computational simplicity, that also showed low ρ^2 values. Nine distinct splits occurred more than twice[6] within the top 300 runs (listed in **Table 9**), which shows each variable's assigned role in the corresponding EB or SB subset. The final column reports the frequency with which each variable appears in SB among these nine high-performing splits. The results suggest that certain EB/SB role *combinations* may represent "sweet spots" that enhance overall model performance. In addition, we can observe that most variables tend to contribute to the model accuracy more effectively when assigned to SB rather than EB (the last column's fraction exceeding 0.5) in the repeatedly high-performing model configurations.

[6] The top 300 subset represents approximately 0.5% of all evaluated instances. If we randomly draw 300 model runs from the pool (4,095 EB/SB splits), the expected number of a specific configuration that appears in the draw is $300/4095 \approx 0.073$ assuming the replacement for simplicity. Denoting this expected number as λ and applying the Poisson distribution $P(X = k) = e^{-\lambda}\lambda^k/k!$, the approximated probability of a split appearing more than twice ($P(X > 2)$) in this 300-draw is approximately 0.00024. In expectation, less than one split (0.98) out of 4,095 would be likely to appear more than twice purely by chance

Table 9 Variable-role assignments (EB vs. SB) for split configurations appearing more than twice among the top 300 ($\rho^2 > 0.446$) model runs

Variable	**Split 25** (3)[a]	**Split 88** (3)	**Split 196** (3)	**Split 246** (3)	**Split 280** (4)	**Split 469** (3)	**Split 641** (3)	**Split 701** (3)	**Split 1445** (3)	**Number of times in SB**
AGE	SB	EB	SB	SB	SB	SB	SB	SB	SB	8/9
MALE	SB	SB	EB	EB	SB	SB	EB	EB	SB	5/9
INCOME	EB	SB	SB	SB	SB	EB	SB	EB	SB	6/9
GA	SB	SB	EB	SB	SB	SB	SB	SB	EB	7/9
PURPOSE	SB	SB	SB	SB	EB	SB	EB	SB	EB	6/9
ORIGIN	SB	SB	SB	SB	EB	SB	SB	SB	EB	7/9
DEST	EB	EB	SB	SB	SB	EB	SB	SB	SB	6/9
FIRST	SB	SB	EB	SB	SB	EB	EB	SB	EB	5/9
TICKET	SB	SB	SB	SB	SB	SB	SB	EB	SB	8/9
PAIDBY	SB	EB	SB	SB	SB	SB	EB	SB	SB	7/9
LUGGAGE	SB	SB	SB	EB	SB	SB	SB	SB	SB	8/9
SEATS	SB	SB	SB	EB	EB	EB	SB	EB	EB	4/9

[a] The number in parentheses shows the number of times the split appears in the top 300 model runs

Second, using the same 61,425 trials (excluding the five $C = 1$ trials), we examined how the configuration parameters interact with the embedding design choice (class-specific vs. shared). **Figure 8** presents violin plots visualizing the distribution of model accuracy across these settings. The results show that class-specific embeddings consistently outperform shared embeddings, as they allow distinct mappings for each latent class (as illustrated in **Figure 7**) enhances the diversity of utility representations across classes, thereby improving the accuracy of choice predictions. In addition, consistent with the trend observed in **Table 5** derived from a fixed manual EB/SB split, the exhaustive grid search further confirms that increasing the number of latent classes generally leads to higher model accuracy. Notably, the best-performing configuration across all 61,430 runs (including the case $|EB| = E = 12$), corresponding to a model of three latent classes with class-specific embedding applied, achieved a test ρ^2 value at 0.482 and a log-likelihood of -1029, significantly higher than any model reported in **Table 6**.

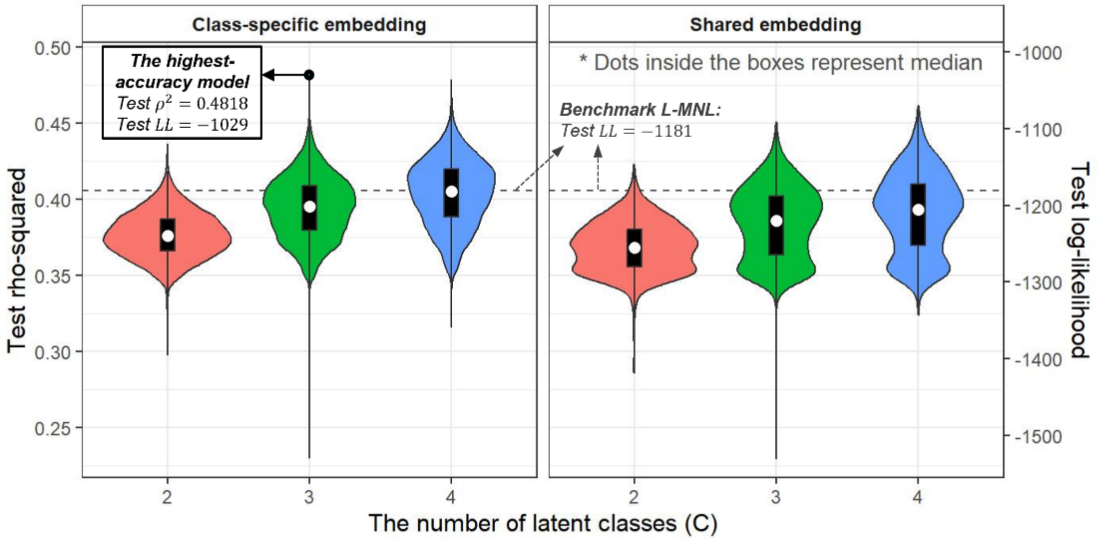


Figure 8 Model accuracy comparison across embedding settings using 61,425 filtered trials of SwissMetro grid search

Figure 9 reclassifies the 61,430 model runs by the number of variables assigned to SB and visualizes the resulting test ρ^2distributions (box plots) by the number of latent classes defined in each configuration, along with the aggregated 95th percentile values (line plot). The extreme cases of NN-LCCM ($E = 0$) and E-MNL ($E = 12$) show the lowest performance, whereas the generalized DCM-SEAL configurations ($E = 1$ to $E = 11$) exhibit substantially higher and more stable accuracy distributions, again proving the synergy of embedding and segmentation. Interestingly, the higher model performance is concentrated around configurations when two to four EB variables (i.e., ten to eight SB variables) are included in the split, suggesting that an optimal property of variable splits may exist where the balance between segmentation and embedding is most effective, which could also vary by the dataset. In addition, even after accounting for differences in the number of runs across split sizes, the overall trend indicates that configurations with more SB variables exhibit greater variability in model performance. Moreover, the interquartile ranges (the height of a box) increase with the number of latent classes among the model runs from the same E value, consistent with the findings in **Table 5**, models with more latent classes defined tend to achieve higher accuracy but also exhibit greater variability.

To further examine the marginal effects of variable-role assignments, we decomposed each of the 61,430 runs into 12 sub-records (one per each non-core categorical variable), preserving only its EB or SB designation and the corresponding test ρ^2. This produced a total of 737,160 marginal observations, which were then grouped into 24 subsets (12 variables × 2 possible roles) to assess the distribution of model performance. **Table 10** reports the 99th-percentile test ρ^2 values for each variable when assigned to SB or EB, along with their differences. Consistent with the trends observed in **Table 9** and **Figure 9**, all variables, except SEATS, achieve higher top-end performance when their role is SB. However, this analysis should be interpreted with caution: it is inherently marginal and does not capture the synergistic effects arising from the joint interaction between segmentation and embedding components within the DCM-SEAL framework. In addition, while this analysis provides useful empirical evidence, the optimal assignment of variables to segmentation and embedding roles remains an open question, warranting future research into automated approaches that leverage learned representations and class membership information from model training.

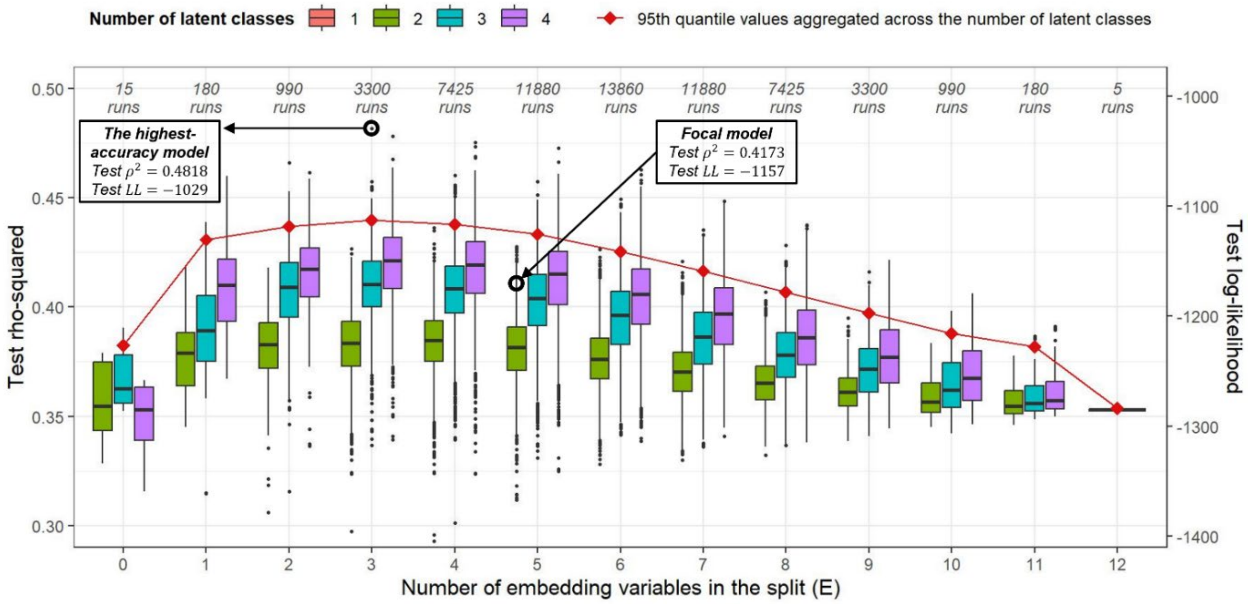


Figure 9 SwissMetro grid search's test ρ^2 distributions by the number of embedding variables and latent-class specifications

Table 10 Marginal performance comparison of each SwissMetro categorical variable when assigned to SB or EB

Variable	99th percentile test ρ^2 when in SB	99th percentile test ρ^2 when in EB	SB-EB (Difference)	Variable	99th percentile test ρ^2 when in SB	99th percentile test ρ^2 when in EB	SB-EB (Difference)
AGE	0.4443	0.4393	0.0051	DEST	0.4450	0.4361	0.0090
MALE	0.4428	0.4408	0.0020	FIRST	0.4428	0.4407	0.0021
INCOME	0.4437	0.4391	0.0045	TICKET	0.4458	0.4309	0.0149
GA	0.4432	0.4403	0.0029	PAIDBY	0.4436	0.4398	0.0038
PURPOSE	0.4431	0.4405	0.0026	LUGGAGE	0.4442	0.4390	0.0052
ORIGIN	0.4442	0.4385	0.0056	SEATS	0.4405	0.4432	-0.0027

6. Conclusion

Building upon and encompassing the previous works on E-MNL (Arkoudi et al., 2023) and NN-LCCM (Baek and Khani, 2025; Han, 2019), this study introduced the Discrete Choice Model with Segmentation and Embedding-Augmented Learning, a unified framework that bridges classical econometric modeling and modern artificial-intelligence-based learning. By jointly integrating 1) a feed-forward deep artificial neural network for latent-class segmentation, 2) class-specific embedding matrices for representing and interpreting high-cardinality categorical variables, and 3) a conventional linear-in-parameter utility formulation, DCM-SEAL captures complex behavioral heterogeneity while preserving core interpretability.

Across both the controlled synthetic experiments and the empirical SwissMetro application, DCM-SEAL consistently outperformed benchmark models in terms of predictive accuracy and behavioral coherence. The synthetic data experiments confirmed that DCM-SEAL could recover known utility structures and choice probabilities with higher prediction accuracy and behavioral alignment than conventional and data-driven counterparts. In the real-world SwissMetro case, the model reproduced plausible value-of-time estimates and distinct latent classes aligned with meaningful behavioral segments, offering an additional opportunity for richer interpretation through class-specific embeddings. These results demonstrate that the proposed framework not only improves quantitative fit but also enhances qualitative aspects, allowing analysts to extract tailored policies, such as subgroup-specific sensitivities to travel time and fare, and modelers to apply the derived insights for downstream applications, such as activity-based simulations.

Beyond performance metrics, DCM-SEAL offers practical advantages for data exploitation and model deployment. It inherits E-MNL's embedding-based dimensionality reduction, enabling variable- and level-specific interpretation across alternatives, while leveraging NN-LCCM's neural membership flexibility for class-level heterogeneity and policy evaluation. The modular architecture and PyTorch-based implementation further facilitate extensibility, allowing transportation researchers and practitioners to adapt the framework to diverse choice contexts with minimal structural modifications.

Nevertheless, several challenges remain, which present promising avenues for future research:

- First, the current implementation assumes a predefined number of latent classes and manually assigned variable roles, unless the time-consuming grid search option is employed. Future studies should explore automated determination of class structure and variable assignments by leveraging information in each model run's learned representations and class membership probabilities. For example, genetic algorithms or combined TPE searches could automatically optimize the variable-role assignments within a fraction of the computational time compared to the grid search. This algorithmic refinement will substantially enhance the operational scalability and practical appeal of the DCM-SEAL framework for public planning agencies.
- Second, while the model partially mitigates endogeneity by isolating correlated variables through segmentation and embedding, further extensions incorporating instrumental-variable or structural-equation components could strengthen its causal interpretability. Alternatively, adopting more flexible structures (e.g., alternative-specific core parameters, nonlinear structures) may also enhance explanatory capacity.
- Third, the robustness and consistency of behavioral indicators, such as marginal rates of substitution and elasticity measures, warrant additional investigation across datasets with varying scales, sampling designs, and temporal coverage.

- Fourth, following the LCCM tradition, methodological frameworks for explanatory, exploratory, or hybrid class definition and interpretation should be revisited and further developed for ML-integrated latent class models.
- Fifth, expanding the hyperparameter search space and adopting additional regularization strategies could help shape model behavior in ways that are more behaviorally consistent and interpretable.
- Finally, computational efficiency and convergence diagnostics for large-scale applications remain active areas for improvement. Although a single model estimation runs in less than two minutes on modern GPU hardware which can also be parallelized[7], the exhaustive grid search across 12,286 variable-role splits required approximately half a month of calculation, making it impractical for routine, short-term planning applications. However, this computational cost represents a research benchmark designed to explore the global split space. For real-world deployment, practitioners can bypass the grid search entirely by utilizing the manual split strategy, which delivers robust and highly interpretable results in a single training run.

In summary, DCM-SEAL represents a step toward a generalized and interpretable ML-integrated DCM paradigm. It provides a methodological bridge between theory-driven econometrics and data-driven intelligence, empowering transportation researchers and planners to extract deeper behavioral insights from increasingly rich mobility data. Future studies building on this foundation may extend its use to other behavioral domains, such as trip generation, technology adoption, and multimodal network assignment, thereby advancing the broader goal of transparent, data-driven, and equitable transportation decision-making.

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All the findings reflect the results obtained from the analysis conducted by the authors and do not represent the sponsors' opinions. Any limitation of the study remains the sole responsibility of the authors.

Declaration of Generative AI and AI-assisted Technologies in the Writing Process

During the preparation of this work the authors used OpenAI GPT-5 in order to improve the readability and language of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

[7] The models were trained on a Linux workstation equipped with an Intel® Xeon® W7-2595X (26 cores) CPU and an NVIDIA RTX™ 5000 Ada Generation (32 GB) GPU. Both processors contributed substantially to the computation, and each could become a bottleneck depending on the task: mini-batching relies heavily on CPU performance for data preparation and transfer, while the GPU predominantly determines the training speed across epochs).

Appendix A. Synthetic Choice Data Generation

Individual and Contextual Attributes

For $N = 12{,}500$ individuals (indexed n), the following four categorical variables are independently drawn as follows:

- $s_1 \sim Multinomial([s_{11} = 0.47, s_{12} = 0.48, s_{13} = 0.05])$
- $s_2 \sim Multinomial([s_{21} = 0.78, s_{22} = 0.07, s_{23} = 0.15])$
- $e_1 \sim Multinomial([e_{11} = 0.47, e_{12} = 0.40, e_{13} = 0.13])$
- $o_1 \sim Multinomial([o_{11} = 0.2, o_{12} = 0.3, o_{13} = 0.4, o_{14} = 0.1])$

e_2 is designed to be dependent on s_2:

- $P(e_2|s_2) = P([e_{21}, e_{22}, e_{23}]|s_2) = \begin{cases} [0.2, 0.6, 0.2], & s_2 = s_{21} \\ [0.4, 0.55, 0.05], & s_2 = s_{22} \\ [0.4, 0.5, 0.1], & s_2 = s_{23} \end{cases}$

Core Attributes

A continuous distance variable is first generated as

- $dist_n = 2 + Exponential\ Distribution(scale = 4)$, rounded to two decimal places.

Then, alternative-specific continuous variables are derived as (subscript a: bus, b: train):

- $x_{1a} = 2 \cdot dist + 0.4 \cdot dist \cdot N(0,1)$
- $x_{1b} = 1.5 \cdot dist + 0.1 \cdot dist \cdot N(0,1)$
- $x_{2a} = 5 + 1 \cdot N(0,1)$
- $x_{2b} = 15 + 2 \cdot N(0,1)$
- $x_{3a} = 1.5 + 0.5 \cdot 1 \cdot (Uniform(0,1) > 0.8)$
- $x_{3b} = 2 + 0.75 \cdot [dist/5]$

All values are rounded to two decimals.

Latent Class Assignment

Per-sample latent-class probabilities are determined from s_1 and s_2, deriving the membership score l_c's and the resulting probability π_c:

- $l_{n1} = \exp(3 \cdot [s_{n1} == s_{12}])$
- $l_{n2} = \exp(3 \cdot [s_{n1} == s_{11}] \cdot [s_{n1}! = s_{21}])$
- $l_{n3} = \exp(3 \cdot (neither\ conditions\ of\ l_1\ nor\ l_2))$
- $\pi_{nc} = l_{nc}/(l_{n1} + l_{n2} + l_{n3})$

The latent class assignment is done with $Multinomial([\pi_{n1}, \pi_{n2}, \pi_{n3}])$

Utility Specification and Choice Simulation

- Given the reference value of $\beta_{x_1} = -0.10$, the other true parameters were derived for them to result in the MRS values in **Table 2**.
- Then, class-specific utilities are computed based on **Equations (15)-(17)**, with the added final error term $\varepsilon \sim N(0, 0.15^2)$
- For each n, using the predetermined class assignment and the utility above, we simulated the mode choice to generate y.

Appendix B. "Best" Hyperparameter Combinations Obtained from Optuna 200-trials that are Used for the Synthetic Dataset Models

Category	Hyperparameter	$\mathcal{M}_4$: E-MNL	$\mathcal{M}_5$: NN-LCCM	$\mathcal{M}_6$: DCM-SEAL
Global	N_E: The number of training epochs	50	50	50
	N_U: The number of mini-batch updates per epoch	40	40	40
	γ: Gradient descent learning rate	0.0646	0.0129	0.0275
Class Membership DNN	N_a ($a \in \{1,2,3,\dots A\}$): the number of nodes of the a-th layer	-	[128, 128]	[128, 64]
	p_d: Dropout probability for DNN weights	-	0.26	0.23
	λ_s: Weight decay for DNN weights	-	0.0032	0.0085
Embedding	λ_e: Weight decay for embedding weights	0.0003	-	0.0021
	Class-specific embedding ("Y" or "N")	-	-	"N"

Appendix C. "Best" Hyperparameter Combinations Obtained from Optuna 200-trials for the SwissMetro Dataset Models

Category	Hyperparameter	E-MNL	NN-LCCM ($C = 2$)	NN-LCCM ($C = 3$)	NN-LCCM ($C = 4$)	DCM-SEAL ($C = 2$)	DCM-SEAL ($C = 3$)	DCM-SEAL ($C = 4$)
Global	N_E	1200	50	60	50	50	75	90
	N_U	1	40	20	40	20	20	10
	γ	0.041	0.013	0.012	0.016	0.012	0.007	0.017
Class Membership DNN	N_a ($a \in \{1,2,3,\dots A\}$)	-	[64, 128, 64]	[128, 64, 128]	[128, 64]	[64, 64, 64]	[128, 64, 256]	[256, 256]
	p_d	-	0.13	0.22	0.33	0.09	0.04	0.27
	λ_s	-	0.0002	0.0022	0.0014	0.0002	0.0008	0.0003
Embedding	λ_e	0.0003	-	-	-	0.0013	0.0001	0.0006
	Class-specific embedding	-	-	-	-	"Y"	"Y"	"Y"

Appendix D. Pairwise unit embedding vector cosine similarity and degree for the SwissMetro focal model

Embedding Vector 1		Embedding Vector 2		Latent Class 1		Latent Class 2	
Variable	Level	Variable	Level	Cosine	Degree	Cosine	Degree
FIRST	*first class*	LUGGAGE	*none*	-0.733	137°	0.050	87°
FIRST	*first class*	LUGGAGE	*one*	0.747	42°	0.937	20°
FIRST	*first class*	LUGGAGE	*several*	0.343	70°	0.045	87°
FIRST	*first class*	SEATS	*airline seat*	0.571	55°	0.640	50°
FIRST	*first class*	TICKET	*one-way*	-0.225	103°	0.170	80°
FIRST	*first class*	TICKET	*two-way*	-0.674	132°	-0.722	136°
FIRST	*first class*	PAIDBY	*employer*	-0.752	139°	0.516	59°
FIRST	*first class*	PAIDBY	*mixed*	0.291	73°	0.715	44°
FIRST	*first class*	PAIDBY	*self*	0.304	72°	0.686	47°
FIRST	*first class*	PAIDBY	*unknown*	-0.362	111°	0.495	60°
LUGGAGE	*none*	LUGGAGE	*one*	-0.097	96°	0.005	90°
LUGGAGE	*none*	LUGGAGE	*several*	-0.749	138°	0.640	50°
LUGGAGE	*one*	LUGGAGE	*several*	-0.195	101°	0.282	74°
LUGGAGE	*none*	SEATS	*airline seat*	-0.972	166°	0.040	88°
LUGGAGE	*one*	SEATS	*airline seat*	-0.117	97°	0.333	71°
LUGGAGE	*several*	SEATS	*airline seat*	0.726	43°	-0.556	124°
TICKET	*one-way*	LUGGAGE	*none*	0.385	67°	-0.530	122°
TICKET	*one-way*	LUGGAGE	*one*	-0.004	90°	-0.105	96°
TICKET	*one-way*	LUGGAGE	*several*	-0.856	149°	-0.970	166°
TICKET	*two-way*	LUGGAGE	*none*	0.967	15°	-0.681	133°
TICKET	*two-way*	LUGGAGE	*one*	-0.056	93°	-0.736	137°
TICKET	*two-way*	LUGGAGE	*several*	-0.886	152°	-0.635	129°
TICKET	*one-way*	SEATS	*airline seat*	-0.291	107°	0.738	42°
TICKET	*two-way*	SEATS	*airline seat*	-0.924	158°	-0.278	106°
TICKET	*one-way*	TICKET	*two-way*	0.604	53°	0.429	65°
TICKET	*one-way*	PAIDBY	*employer*	0.767	40°	0.585	54°
TICKET	*one-way*	PAIDBY	*mixed*	0.863	30°	0.669	48°
TICKET	*one-way*	PAIDBY	*self*	0.103	84°	-0.560	124°
TICKET	*one-way*	PAIDBY	*unknown*	-0.806	144°	0.708	45°
TICKET	*two-way*	PAIDBY	*employer*	0.930	22°	-0.371	112°
TICKET	*two-way*	PAIDBY	*mixed*	0.295	73°	-0.377	112°
TICKET	*two-way*	PAIDBY	*self*	-0.661	131°	-0.977	168°
TICKET	*two-way*	PAIDBY	*unknown*	-0.294	107°	0.238	76°
PAIDBY	*employer*	LUGGAGE	*none*	0.844	32°	0.329	71°
PAIDBY	*employer*	LUGGAGE	*one*	-0.304	108°	0.193	79°
PAIDBY	*employer*	LUGGAGE	*several*	-0.867	150°	-0.398	113°
PAIDBY	*mixed*	LUGGAGE	*none*	0.056	87°	0.086	85°
PAIDBY	*mixed*	LUGGAGE	*one*	0.433	64°	0.426	65°
PAIDBY	*mixed*	LUGGAGE	*several*	-0.704	135°	-0.471	118°
PAIDBY	*self*	LUGGAGE	*none*	-0.784	142°	0.608	53°
PAIDBY	*self*	LUGGAGE	*one*	-0.345	110°	0.774	39°
PAIDBY	*self*	LUGGAGE	*several*	0.425	65°	0.740	42°
PAIDBY	*unknown*	LUGGAGE	*none*	-0.068	94°	-0.819	145°
PAIDBY	*unknown*	LUGGAGE	*one*	-0.551	123°	0.428	65°
PAIDBY	*unknown*	LUGGAGE	*several*	0.702	45°	-0.674	132°
PAIDBY	*employer*	SEATS	*airline seat*	-0.726	137°	0.948	19°
PAIDBY	*mixed*	SEATS	*airline seat*	-0.061	93°	0.994	6°
PAIDBY	*self*	SEATS	*airline seat*	0.897	26°	0.102	84°
PAIDBY	*unknown*	SEATS	*airline seat*	0.105	84°	0.468	62°
PAIDBY	*employer*	PAIDBY	*mixed*	0.386	67°	0.943	19°
PAIDBY	*employer*	PAIDBY	*self*	-0.346	110°	0.172	80°
PAIDBY	*employer*	PAIDBY	*unknown*	-0.334	110°	0.163	81°
PAIDBY	*mixed*	PAIDBY	*self*	0.176	80°	0.210	78°
PAIDBY	*mixed*	PAIDBY	*unknown*	-0.990	172°	0.454	63°
PAIDBY	*self*	PAIDBY	*unknown*	-0.084	95°	-0.237	104°

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